

A Machine Learning Approach to Forecasting Wind Power Generation in the Netherlands

Final Project Report

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<https://github.com/benrodion/Watts-next>

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1. Introduction

2024 was the hottest year in recorded history, with every month setting new temperature records (IEA, 2025). This rise in temperatures was in no small part driven by the energy sector, which accounts for three quarters of global CO₂-emissions (Kiunke et al., 2022). Therefore, the green energy transition is one of the most important levers in the fight against climate change and must be at the core of all measures. At the forefront of this transition is wind power, which is not only the second largest source of renewable energy in terms of installed capacity (IEA, 2025), but also the most cost-competitive (Kiunke et al., 2022). However, the increasing deployment of wind power also poses significant challenges to the energy system. Unlike fossil fuels, wind power is highly volatile, which is why the more wind power there is, the more difficult it becomes to balance the grid load.

The resulting grid instability leads to undesired outcomes, chief among them curtailment and redispatch. Curtailment refers to the shutting down of wind turbines in energy producing regions when they produce more than the grid can absorb and transport, resulting in a waste of electricity. Since lacking grid capacity and curtailment prevent wind energy from meeting energy demand, redispatch i.e. the ramping up of fossil power plants in regions with high energy consumption, is utilised instead to supply the required energy. In recent years, with the expansion of wind energy, the problem of curtailment has also grown (Chen et al., 2022; Synertics, 2024). While the vicissitude of wind power cannot be changed, it is possible to improve demand flexibility in the long term with accurate predictions of wind power generation, thereby improving grid management. In light of the challenge curtailment poses to the further deployment of wind power and thus the energy transition as a whole (Synertics, 2024), the importance of such predictions has grown.

The issue outlined here is relevant from a policy perspective, as well as from an economic and machine learning perspective. From a policy viewpoint, improving grid management and reducing curtailment is vital for reaching key policy goals in the context of the energy transition, especially those pertaining to the ramp-up of wind power capacity. For example in Germany and China, two of the global leaders in wind power, curtailment is already a significant issue at the current levels of installed wind power capacity. This makes it difficult to fulfill their ambitious targets, as a further expansion of wind power would not necessarily improve the supply of green energy in either country but rather increase grid congestion and drive up costs (Chen et al., 2022; Synertics, 2024).

From an economic angle, curtailment is simply inefficient because it amounts to wasting sustainable and affordable green energy while at the same time deploying coal- and gas-fired power plants in energy consumption hotspots (CleanEnergyWire, 2024; Joas, 2023).

The costs of this are tremendous and in Germany alone amounted to 3.1 billion EUR in 2023 (Der Spiegel, 2024). Lastly, the problem presented here is interesting from a machine learning perspective, as it is essentially a matter of making accurate predictions based on complex time series data. As will be demonstrated in the exploratory data analysis (EDA), this is an intricate task requiring a deep understanding of the data's structure.

In our machine learning project, we take on this task by developing multiple models to predict future wind power generation based on past generation as well as other predictors. The remainder of the paper is structured as follows: first we translate the problem of predicting wind power generation into a machine learning problem in line with Murphy (2022) and contextualise with papers similar to this one. We then describe the data collection process as well as the data's structure in the exploratory data analysis (EDA) and its challenging nature. Following the EDA, we address the analytical challenges of our data, before introducing the different models we ran and presenting and comparing their results. We embed the results in a discussion based on which we outline avenues for future research before summarising our key findings.

2. Energy Production Forecasting as a Machine Learning Problem

Following Murphy (2022), a well-posed machine learning problem consists of a task T , a performance measure P and an experience E , the goal being that a computer program improves its P in T based on E (Mitchell, 1997). Applying this to our project, we are dealing with a supervised learning problem, because we are trying to forecast future energy production from offshore and onshore windparks, respectively, based on a set of predictors.

Put in abstract terms, the task T therefore is for our models to learn the mapping of the input features to the label. In this context, the experience E is given by the training data, which is a set of feature-label-pairs used to train the models. The performance measure we use to compare the prowess of our models mainly is the RMSE, i.e. the average squared discrepancy between the true value of the label and the value predicted by our models (James et al., 2023; Murphy, 2022).

3. Data Collection

3.1. Wind Energy Generation

The data on offshore and onshore wind energy production were retrieved from the National Energy Dashboard (Nationaal Energie Dashboard, 2025). NED is a platform containing data on all energy production from electricity and gas in the Netherlands from 2017 onwards. It is intended as a resource for both researchers and industry to study the energy transition in the Netherlands and more broadly to gain insight into the contribution of various energy production methods towards the total energy mix.

The NED API allows users to select (amongst other things) the type of energy production, granularity, location (i.e. provinces, Netherlands, offshore) and the desired time window. We requested data on offshore and onshore wind energy with the location on offshore and the Netherlands respectively and the granularity set to hours. Since the API only allows individual requests to retrieve a limited amount of data points, we looped through a list of lists with different date combinations to gain all the data between January 1st, 2017, and December 31st, 2024.

The requested data consists of six key features (see table 3.1). As explained by the NED definition page, multiplying capacity and percentage produces a feature with the total amount of installed capacity at the time of production. In the end, we did not utilise this feature as its values were nearly identical to volume and could not possibly correspond with the actual amount of installed capacity in the Netherlands.

Table 3.1: Dataframe energy observations from NED, identical for offshore and onshore

Column Name	Description
YYYYMMDD	Date of observation (YearMonthDay)
Volume	Volume wind energy produced
Percentage	% of capacity employed during generation
Emission	Amount of emission CO2 in kg
Emission factor	Emissions produced by wind energy generation, zero in each row as renewable energy source
Capacity	Total capacity installed onshore/offshore

3.2. Weather Data

Weather data were retrieved from KNMI, which is The Royal Netherlands Meteorological Institute (KNMI), that provides information on the meteorological indicators of different time granularity. For the purpose of our project, we requested the data from the platform's web servers as [suggested by the developers](#) (the initial Bash code provided by them was 'translated' to Python).

We specified the stations and required variables in our request whilst retrieving the data. The codebook can be consulted on the KNMI [user-friendly data requesting interface](#). The KNMI server was only capable of returning data for one station at a time since we were extracting data for a large timeframe January 1st 2017 to December 31st 2024. We hence retrieved data for each station and then combined the data. In the end, we got a dataframe that contains:

Table 3.2: Dataframe weather observations from meteorological stations

Column Name	Description
STN	Station number
YYYYMMDD	Date of observation (YearMonthDay)
HH	Hour of observation (from 1 to 24)
DD	Wind direction in degrees, averaged over the last 10 minutes of the past hour (360 = north, 90 = east, 180 = south, 270 = west, 0 = calm, 990 = variable)
FH	Hourly average wind speed (in 0.1 m/s)
FF	Wind speed averaged over the last 10 minutes of the past hour (in 0.1 m/s)
P	Air pressure at sea level (in 0.1 hPa) during observation
U	Relative humidity (in %) at a height of 1.5 m during observation

To reduce the amount of memory space taken up by a dataset consisting of 3.2+ mln rows, we adjusted the data types for most columns. Additionally, before calculating the averages for the offshore and onshore stations, we decided to remove 22K rows which contained "990" as value for wind direction. We manually categorized the meteorological stations into either off- or onshore (see figure 3.1), allowing us to conduct a separate analysis for both location specifications respectively.

Figure 3.1: Netherlands onshore and offshore wind station map



An interactive map displaying the locations of these offshore stations is available on the [project map website](#).

In the end, we merged the wind generation data with their corresponding weather data which resulted in two data frames of length 70128. This number corresponds to the number of hours between January 1st, 2017, and December 31st, 2024. These two dataframes formed the basis of our analysis and were used by all models.

3.3. Price Data

Initially, we sought to retrieve data on electricity prices from the Dutch Central Bureau of Statistics. However, their database only includes monthly average prices, whilst we required an hourly granularity. We resorted to downloading data from the website Jeroen.nl which is a company that monitors the dynamic electricity prices for consumers with e.g. home batteries, solar panels and electric cars. Consumers can use the website's dashboard to determine for instance the optimal solar production and charge moments for their appliances. These dynamic electricity prices are the best approximation to real-time electricity prices in the Netherlands we were able to acquire.

The Dutch power market is an extremely complex system that is highly integrated with the European system. We were unable to include future, day-ahead and intraday prices as features in the models as these were too difficult to access in the desired format through ENTSO-E.

4. EDA

4.1. Dataset overview

We analysed two hourly time-series - onshore and offshore wind production - from January 1st 2017 to December 31st 2024. Each data frame contains 70128 consecutive hours. Fourteen columns describe every hour: two columns for generated energy volume, three aggregates for wind speed and direction, respectively, two additional weather variables for humidity and air pressure as well a column for emissions. The remaining columns contain data on the date and hour. There are no missing values. Since some columns were duplicates, they were removed ahead of analysis.

Table 4.1: Overview of Variables

Variable	year_mon_day	hour	wind_dir_avg_10	windspeed_h_avg	wind_speed_avg_10	air_pressure	humidity
Explanation	Temporal variable containing year, month and day as YYYYMMDD	The hour of the day	Indicates the hourly average of wind direction as measured on a circle, ranges from 0-360	Hourly average wind speed (in 0.1 m/s)	Wind speed averaged over the last 10 minutes of the past hour (in 0.1 m/s)	Air pressure at sea level (in 0.1 hPa) during observation	Relative humidity (in %) at a height of 1.5 m during observation

Table 4.2: Overview of Variables cont.

Variable	full_datetime	capacity	volume	percentage	emission	emission_factor	correct_days
Explanation	Date and hour of the day as YYYY-MM-DD-HH	Total installed capacity	Volume of produced wind energy	% of capacity employed during generation	Amount of emission CO2 in kg	Emission factor used for this energy carrier in kg/kWh	Duplicate of full_datetime

4.2. Summary Statistics

Table 4.3 contrasts average meteorological and production metrics for on- and offshore wind production. The most pronounced gap lies in mean wind speed: 66.3 m/s at sea versus 41.7 m/s on land - about a 60 % increase offshore. This stronger wind is mirrored by a higher average energy output for offshore turbines. In contrast, atmospheric pressure (~10 150 Pa) and relative humidity (~80 %) are nearly identical for both location specifications.

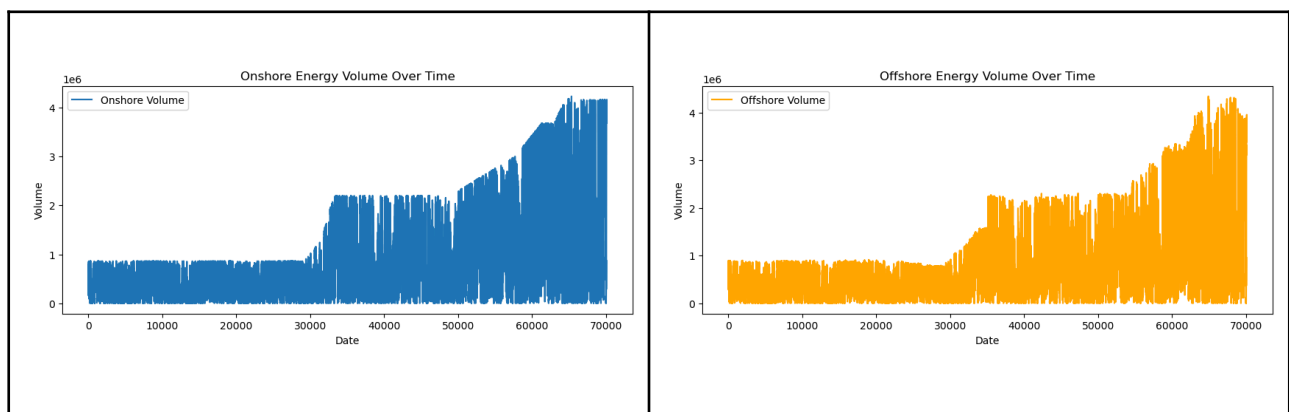
Table 4.3: Main Descriptive Facts

Metric	On-shore	Off-shore	Comment
Mean wind speed (m/s)	41.7	66.3	≈ 60 % faster at sea
Mean volume (MWh)	9.1×10^5	8.3×10^5	More production at sea
Mean air pressure (Pa)	10 153	10 147	Almost identical
Mean relative humidity (%)	80	81	Almost identical

4.3. Temporal Structure

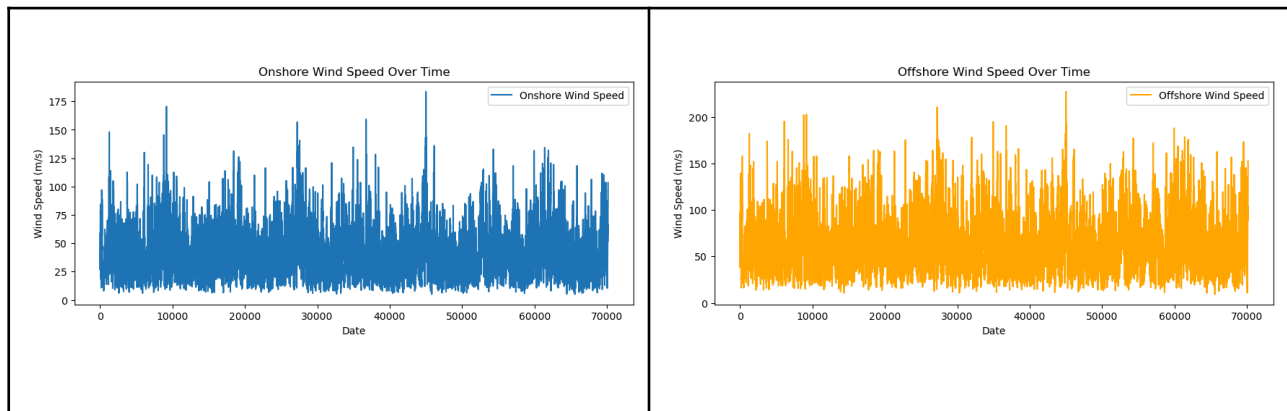
Our EDA reveals a clear upward trend in our target variable over time, which is characterized by stepwise increases, presumably corresponding to the expansion of wind power capacity in the Netherlands. The trend contained in the data speaks to its complexity and points to potential challenges for the analysis, as a trend implies non-stationarity, meaning that statistical properties (mean, variance, covariance) of key variables change over time.

Graph 4.1: Energy Volume over Time



The seasonality analysis points in a similar direction. As graph 2 shows, there is strong annual seasonality in wind speed with a peak during winter, again implying that our data is non-stationary.

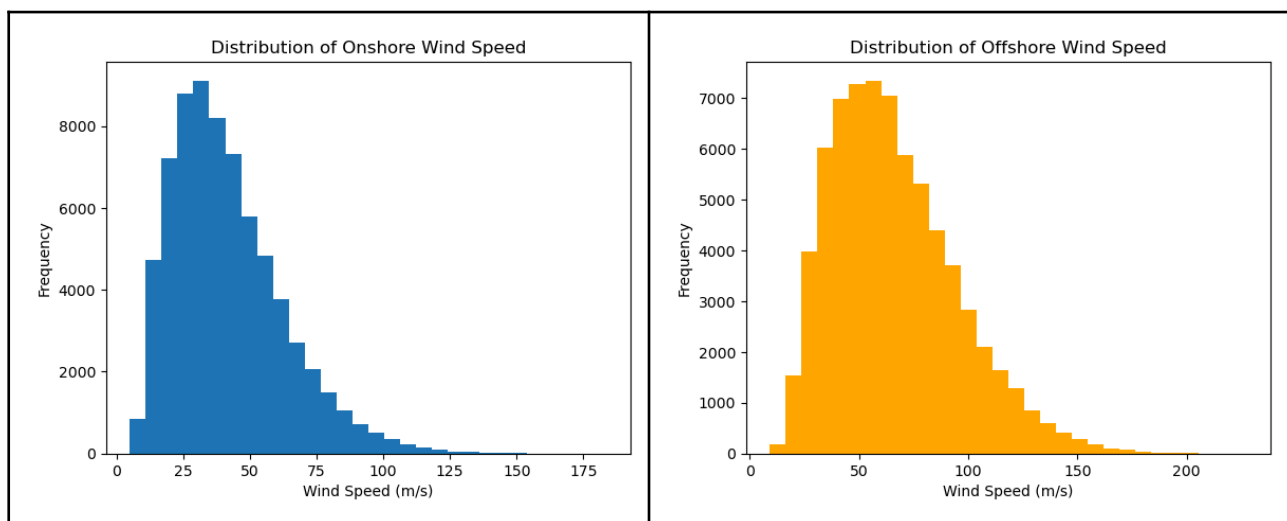
Graph 4.2: Wind Speed over Time



4.4. Distributions

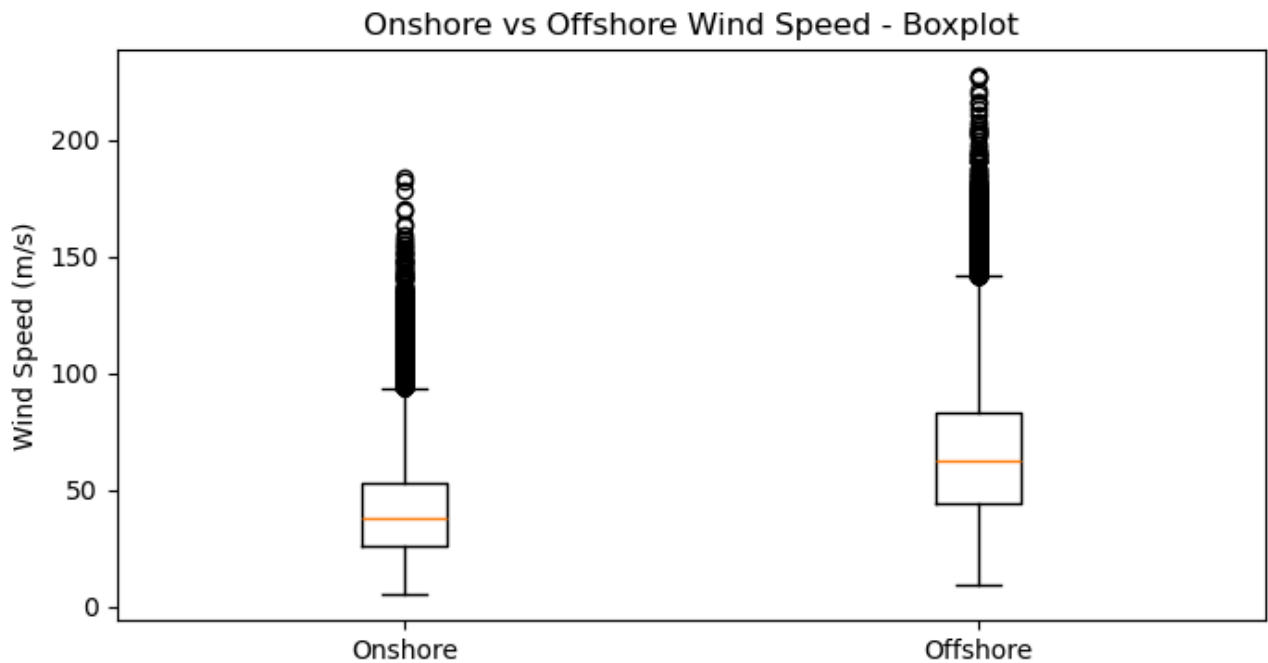
The distributions for both onshore and offshore wind speed are right-skewed and show that offshore wind speed generally is a lot higher than onshore wind speed, underlining the findings from the descriptive statistics. For onshore wind power plants, 50% of the data points cluster around a wind speed of 26-52 m/s, whereas for offshore they cluster around a wind speed of 44-83 m/s.

Graph 4.3: Distribution of wind speed



These patterns are confirmed by boxplots, showing that wind speed is generally higher for offshore wind turbines, but also has a greater variance. Furthermore, they reveal that extreme gusts reach 228 m/s at sea compared to 185 m/s on land.

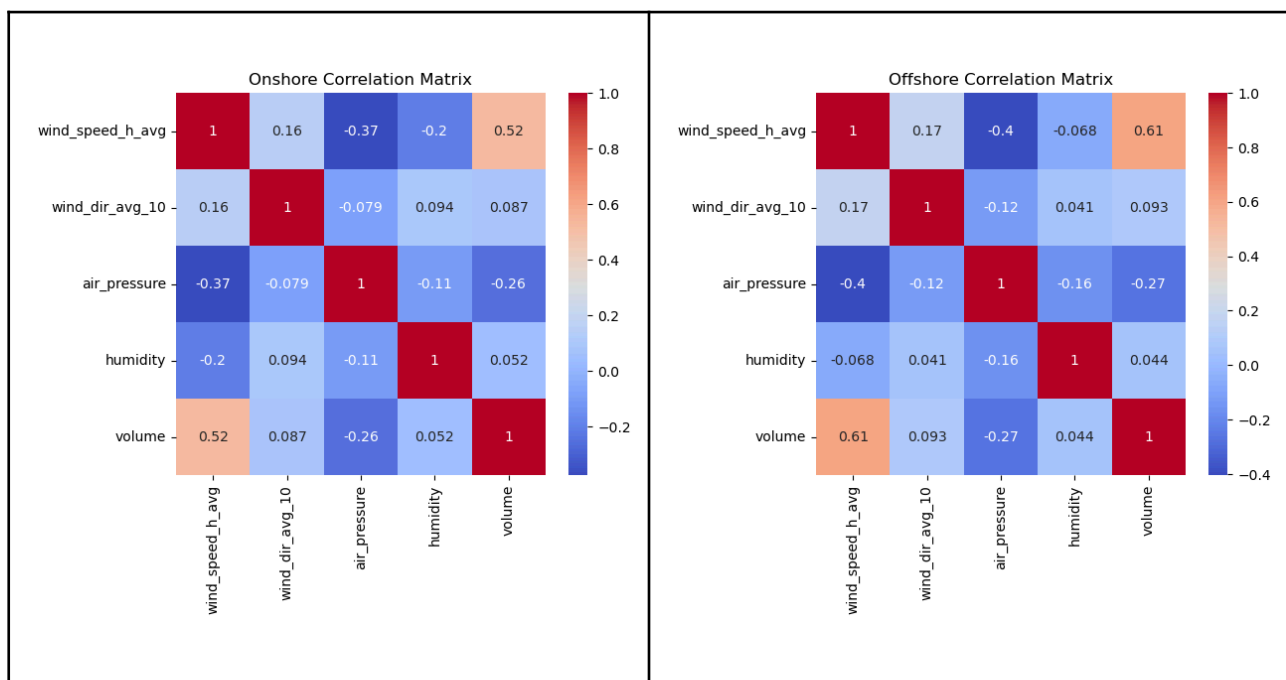
Graph 4.4: Wind speed - boxplot



4.5. Relationships between variables

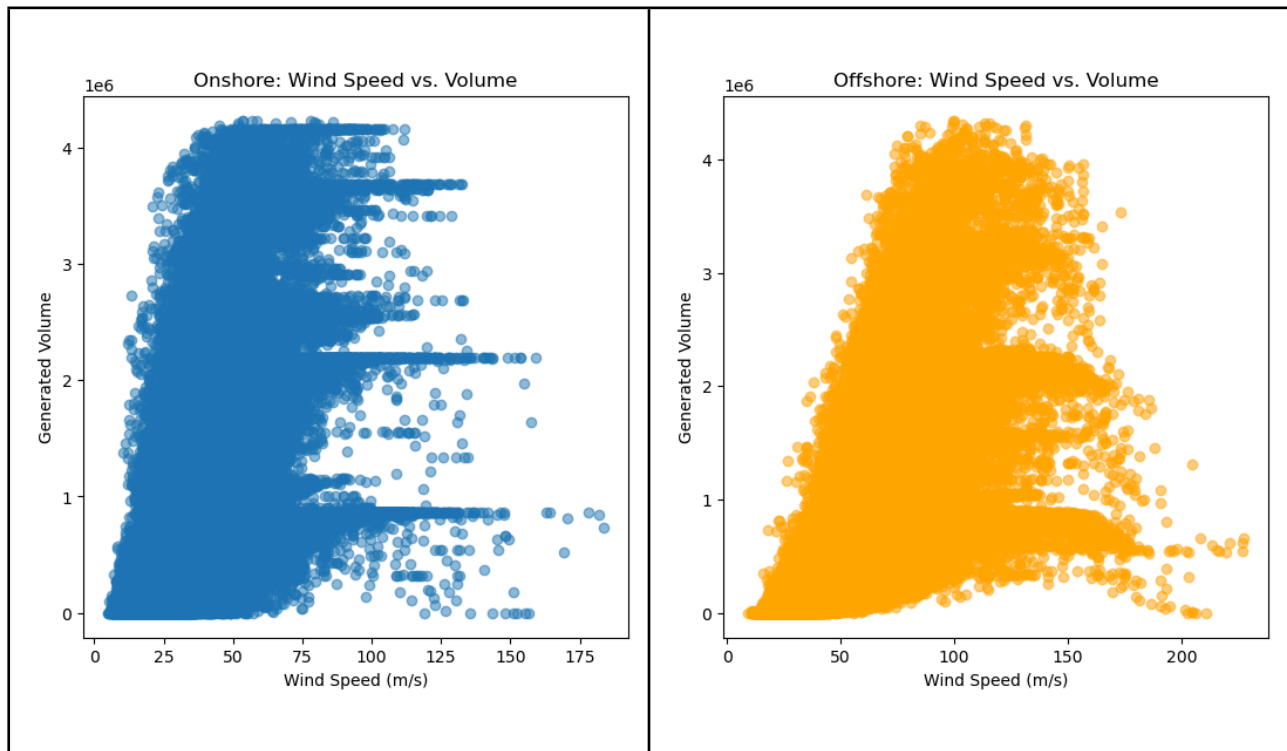
Unsurprising, the variable we included for wind speed strongly correlates with generation volume, with values of 0.52 for onshore and 0.61 for offshore plants. Regarding the other weather inputs, the correlation with the target variable is much weaker, with the exception of air pressure. Air pressure is also correlated with wind speed and through this presumably correlates with volume.

Graph 4.5: Correlation Matrices



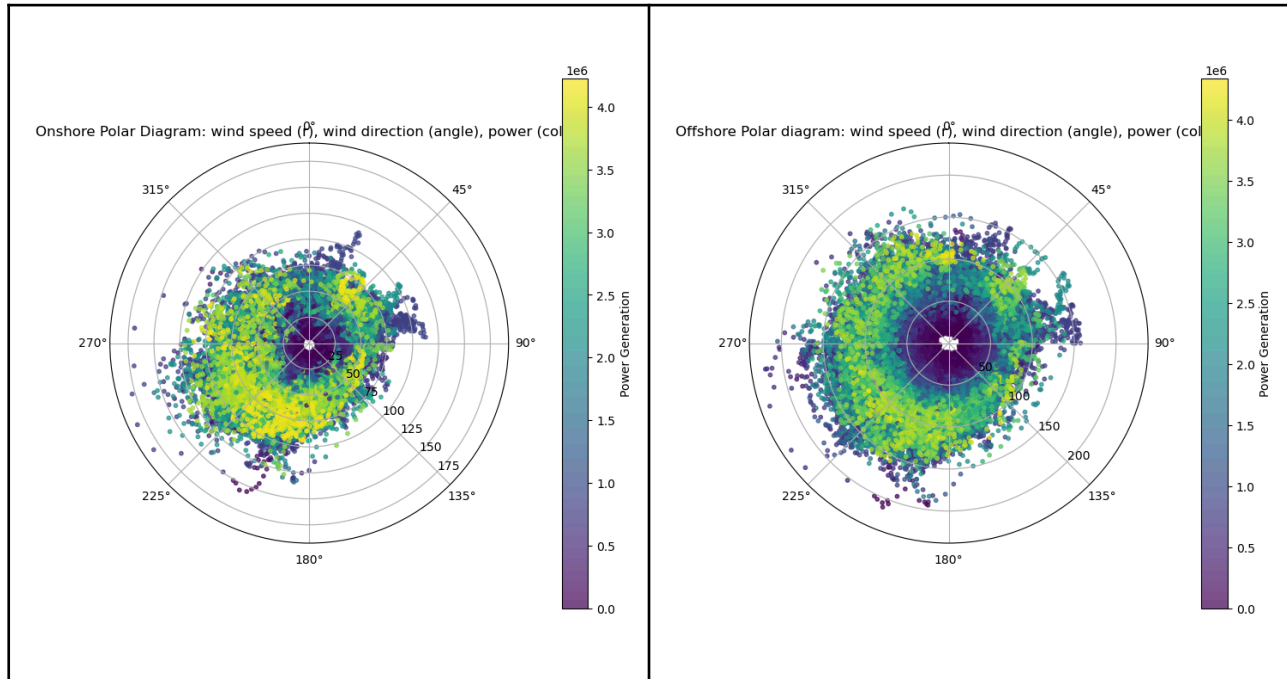
A closer look at the relationship between wind speed and energy generation volume yields an interesting pattern. Although a positive correlation is identifiable, there is a lot of noise in the data. As we can see, the highest wind speeds do not correspond to the highest generated volume, because of the so-called cut out; if wind speed exceeds a certain limit, turbines are shut down to prevent damage. Furthermore, turbines are often deactivated as part of managing the grid, which also impacts the strength of the correlation between wind speed and wind energy generation.

Graph 4.6: Wind Speed vs Volume



The polar diagrams reveal differences in wind direction between onshore and offshore turbines. While the former rely chiefly on stronger south-westerly winds, the latter are exposed to uniformly higher, almost omnidirectional winds.

Graph 4.7: Polar Diagrams



4.6. Implications for modelling

1. The series combines trend, seasonality and non-constant variance. Simple stationary autoregressive models will fail without transformations (e.g. log or differencing).
2. Wind speed is the key predictor; we must keep other variables as fine-tuning features.
3. Different models perform differently on multiple horizons, so we aggregate data to hourly, daily, weekly and monthly views and split each into train/test windows. This supports both strong predictive algorithms (e.g. gradient-boosted trees) and seasonal models (SARIMAX, Holt-Winters).

In short, stronger and more volatile winds at sea drive larger but less predictable output; land-based production is steadier. Any successful model must capture this contrast, respect the yearly rhythm and cope with changing variance.

5. Addressing the challenges of time series datasets

Our data pose two problems for modelling: non-stationarity and high granularity. This chapter is dedicated to understanding these challenges and how we tackle them.

5.1. First challenge: non-stationary data with trends and seasonality

Given the nature of our data, we hypothesized that the hourly wind energy production dynamics of offshore and onshore wind turbines were non-stationary. Non-stationary datasets have the following features:

- (1) non-constant mean due to trends in the data
- (2) changing autocorrelation structure due to seasonality
- (3) non-constant variance of residuals (heteroscedasticity)

To test our hypothesis of non-stationarity, we performed a seasonal decomposition of both datasets. The decomposition indicated that both the onshore and offshore data are non-stationary. The decomposition process revealed similar patterns in trends, seasonality, and residuals. Both datasets exhibited the following features:

- (1) an upward trend with strong growth in 2020-2021 and post 2023
- (2) strong annual seasonality
- (3) increasing volatility in residuals after 2021

These features occur due to the nature and context of our data. First, the Netherlands experienced significant expansion in wind energy capacity after 2020, contributing to the observed upward trend. Second, the annual seasonality in wind energy production stems from its strong dependence on wind speed, which follows seasonal patterns with stronger winds in winter and milder winds in summer. Third, the increased volatility likely results from the growing installed capacity of wind turbines, which amplifies the system's sensitivity to weather fluctuations.

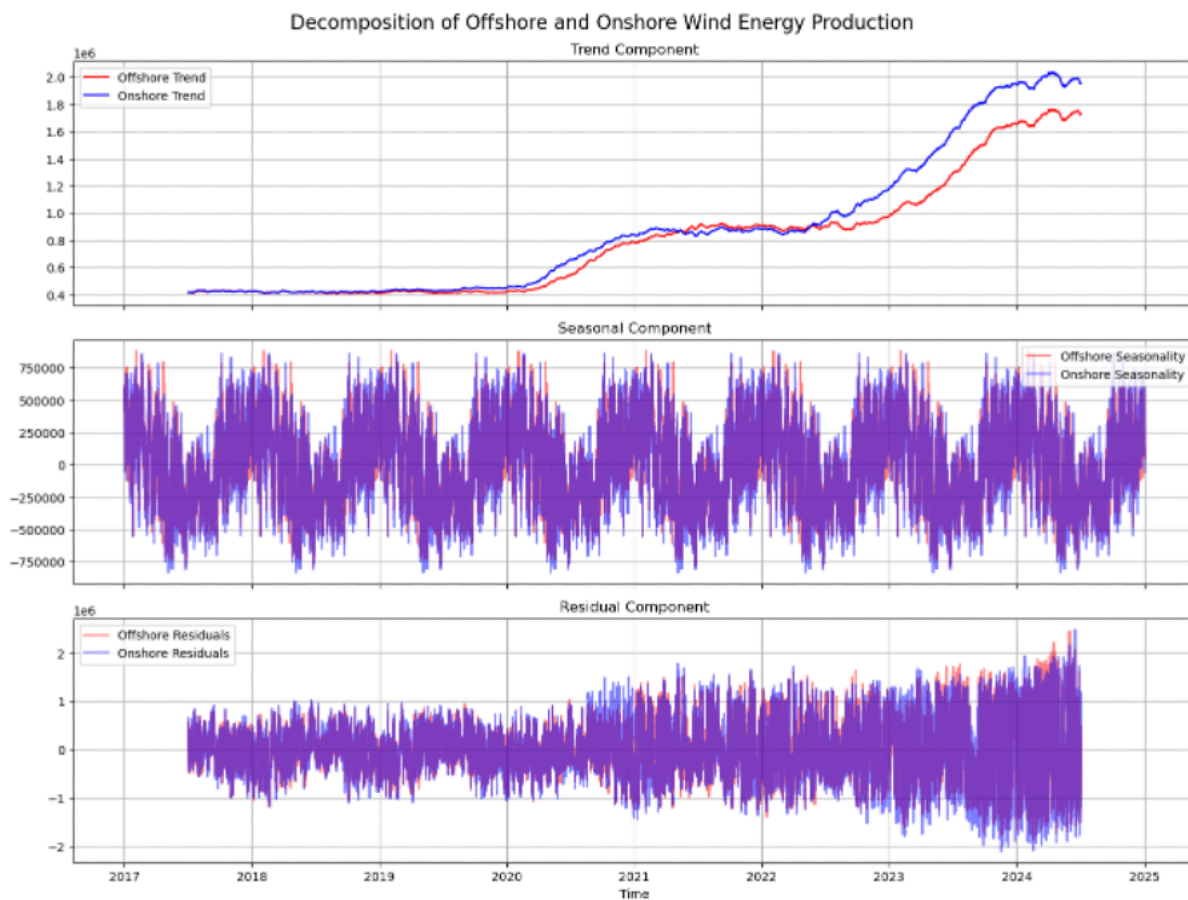
Given the hourly granularity of our data, we also conducted a periodogram analysis to confirm whether annual seasonality has the most significant impact on our data. The results of the periodogram affirmed the dominant influence of annual cycles. Additionally, we confirmed the heteroscedasticity of residuals using the widely recognised Breusch-Pagan test.¹

¹ Breusch-Pagan test for the offshore dataset: 24297.2, p-value: 0.0, which indicates strong heteroscedasticity. Breusch-Pagan test for the onshore dataset: 27650.9, p-value: 0.0, which indicates strong heteroscedasticity.

Finally, we assessed overall stationarity using two complementary tests: ADF (Augmented Dickey-Fuller) and KPSS (Kwiatkowski–Phillips–Schmidt–Shin). The results we got – with ADF suggesting stationarity while KPSS indicating non-stationarity – are not unusual for time series with strong deterministic trends and seasonal components.² Given the presence of upward trends, annual seasonality, and heteroscedasticity in both datasets, we proceeded with modelling approaches designed to address non-stationary features.

Graph 5.1: Seasonal Decomposition

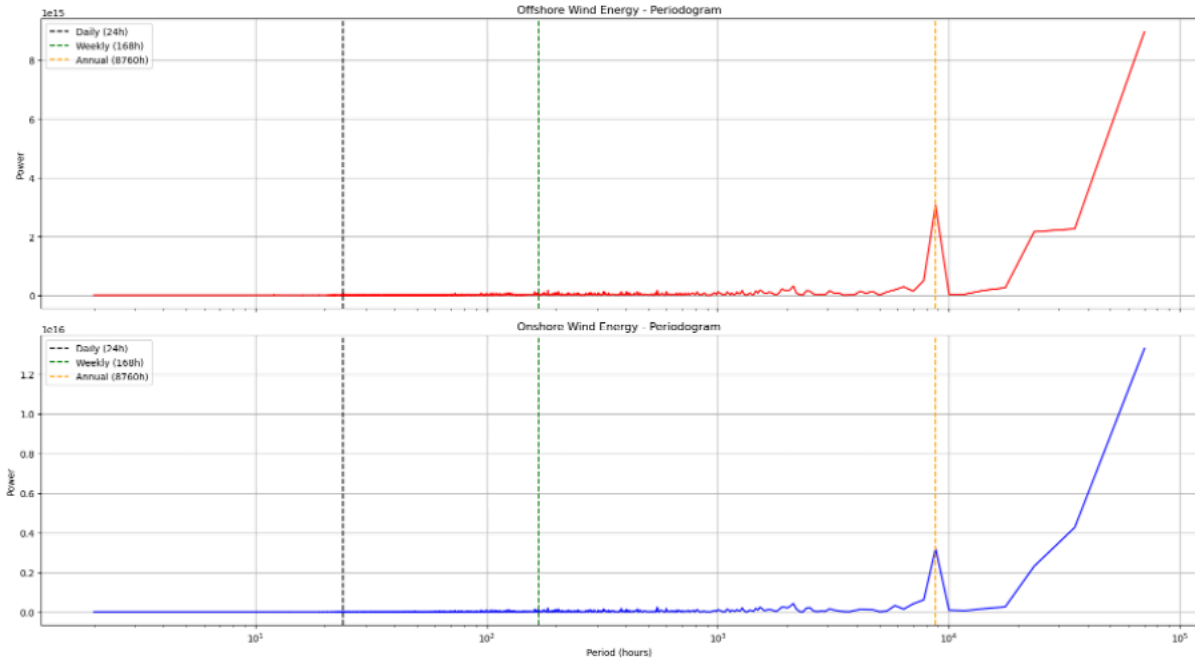
The violet area indicates a shared area for both offshore and onshore data.



² The ADF test sometimes indicates stationarity when the datasets show strong upward trends and regular seasonal patterns. This occurs because the ADF test primarily checks for stochastic non-stationarity and can misinterpret trends and seasonal patterns as forms of stationarity, especially when these components create predictable annual fluctuations. This is most likely the case with our dataset.

Graph 5.2: Periodograms

The orange line (8760 h) indicates strong annual seasonality.



5.2. Second challenge: high granularity of data

Both datasets for offshore and onshore wind energy production feature exceptionally high temporal resolution: 70,128 hourly measurements spanning from January 1st, 2017, to December 31st, 2024. This high granularity, combined with growth trends and dominant annual seasonality, presented significant modelling challenges.

Rather than attempting to directly model hourly predictions, we implemented a multi-granularity approach. This decision allowed us to address the complexity of the data by working across different temporal scales. We created 4 distinct versions of the dataset through temporal aggregation:

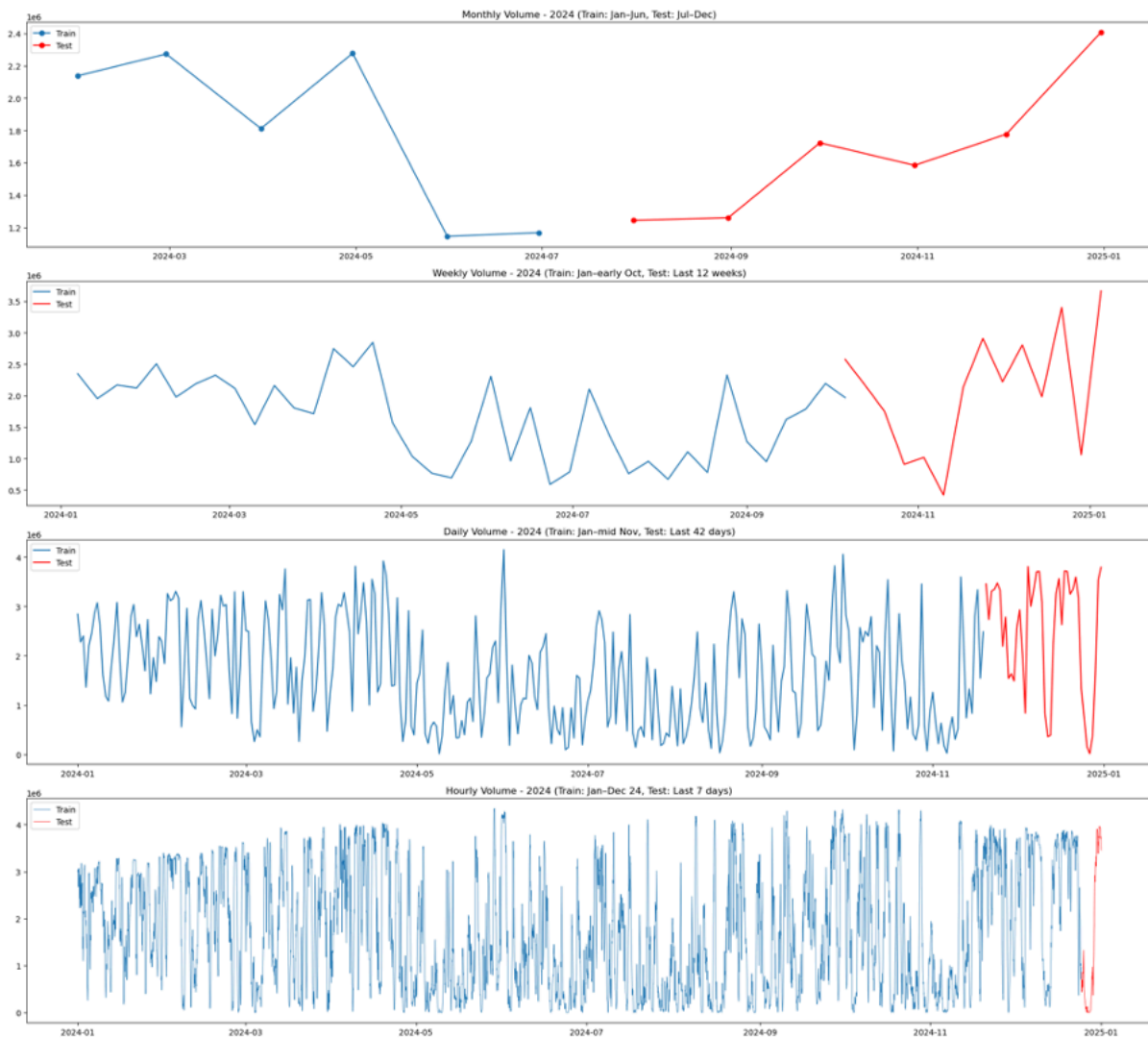
- (1) monthly data (96 values)
- (2) weekly data (418 values)
- (3) daily data (2,922 values)
- (4) hourly data (original 70,128 values)

Each of these temporal resolutions was then split into appropriate train and test sets. This deliberate multi-granularity strategy enabled us to build and refine our predictive understanding progressively, starting with the more manageable monthly predictions before tackling the substantially more complex hourly forecasting challenge.

Table 5.1: Train and test sets for datasets of different granularity

Time granularity	Prediction period	Train set	Test set
Monthly	6 months	Jan 2017 – Jun 2024	Jul 2024 – Dec 2024
Weekly	12 weeks	Jan 2017 – Sep 2024	Oct 2024 – Dec 2024
Daily	42 days	Jan 2017 – Nov 2024	Nov 2024 – Dec 2024
Hourly	168 hours (7 days)	Jan 2017 – Dec 2024	Dec 2024 (7 days)

Graph 5.3: Train and Test-Sets Used in the Models Offshore Data



6. Model Specification

A wide variety of models was employed to predict on- and offshore wind energy production. We opted for three baseline models (Holt-Winters, SARIMAX & Ensemble Model (Polynomial + Random Forest Regression)) and three complex models (XGBoost, Chronos, Prophet). The distinction between baseline and complex allows us to establish whether simpler models are capable of explaining our highly complex data or if the challenges detailed in the previous section are too severe for ‘simple’ regression models. Our diversity in model selection also enabled group members to investigate different solutions to the data challenges. We also attempted to develop an LSTM-model which due to severe performance issues is not included in the report.

Therefore, our project presents a holistic set of models and model specifications to predict on- and offshore wind energy production in the Netherlands. This section provides an overview for each included model regarding its computational approach, parameters and variable selection.

6.1. Baseline Models

6.1.1. Holt-Winters and SARIMAX

We started with two standard baseline models for time series forecasting: (1) the Holt-Winters exponential smoothing model and (2) the SARIMAX (Seasonal AutoRegressive Integrated Moving Average with eXogenous variables) model.

The Holt-Winters model often serves as the most basic model for predicting non-stationary time series data. It uses exponential smoothing, which exponentially reduces the weight of older observations, and applies this to both trend and seasonal components. The model also demands little computational effort because of its straightforward structure.

However, the Holt-Winters model has several notable limitations: it cannot incorporate external explanatory variables, frequently struggles with non-linear trends, and often shows significant performance degradation as data granularity increases. Despite these drawbacks, we considered the Holt-Winters model as a good starting point for our modelling endeavour.

For our implementation, we configured the Holt-Winters model with:

1. Additive (linear) trend and additive (annual) seasonality
2. Different seasonal periods (12 for monthly data, 52 for weekly data, 365 for daily data, and 8760 for hourly data)

The results of the Holt-Winters model aligned with our expectations. This model underperformed compared to other baseline models and showed particularly poor quality for high-granularity data (daily and hourly observations). Please consult appendices A-D for graphs of the modelling results for the different granularities.

The SARIMAX model, an extension of the traditional ARIMA framework, proved particularly suitable for our energy forecasting task for several reasons. First, it addresses our data challenges by applying differencing to achieve stationarity and incorporating seasonal components. Second, unlike Holt-Winters, SARIMAX accommodates external explanatory variables (such as wind speed and direction), which are critical for our modelling task. Third, both SARIMAX and its simpler variant SARIMA, are widely recognised in the literature as strong baseline options for energy time series forecasting (Elamin and Fukushima 2018, Dubey et al. 2021).

A key limitation of SARIMAX is its computational constraints when handling highly granular data with extensive seasonal patterns. Due to these limitations, we did not run the model for hourly data.

For our implementation, we configured the SARIMAX model with:

1. a seasonal autoregressive term to predict the current value
2. both trend and seasonal differencing to ensure stationarity
3. different fixed seasonal periods corresponding to the data granularity: 12 for monthly data, 52 for weekly data, and 365 for daily data

We enhanced the model by incorporating scaled and standardised exogenous variables, including wind speed and direction.

Surprisingly, the SARIMAX model delivered strong results across both offshore and onshore datasets at monthly, weekly, and daily granularities, outperforming more complex models. This success can be attributed to SARIMAX's specific design for time series characteristics (autocorrelation, trends, etc.) and its approach to non-stationarity through differencing. Additionally, SARIMAX's explicit handling of seasonality at fixed intervals helps avoid the overfitting issues that may have affected more complex models. This case illustrates Occam's razor in machine learning: simpler models can sometimes forecast complex data more accurately than their more sophisticated counterparts.

6.1.2. Ensemble model (Polynomial plus Random Forest Regressor)

Our range of baseline models also includes some classical regression models in the temporal context. We employed the `sktime` Python module which adapts useful tools from the `sklearn` toolbox for time series forecasting purposes, examples include but are not limited to data splitting, setting forecast windows and performing grid search along with cross-validation. Limited computational capacity constrained our ability to utilise all these options and we could hence not perform automated grid search for ensemble models. Instead, we resorted to fine-tuning the parameters manually; i.e. rather than considering all possible parameter combinations, we iteratively changed a single parameter, keeping others fixed until we found an optimal solution.

The ensemble models were trained with all exogenous weather variables. The model types are selected with the aim to reveal different characteristics of the data:

1. Polynomial models reflect global fluctuations of the target variable through time.
2. Random forest regressors with window lengths of 2 for monthly data (due to small total number of observations) and 5 for all other granularities. We employed it to capture the most recent, short-term dynamics of the target based on previous several observations.

In daily and hourly predictions, we ended up assigning 95% weight of the final estimate to the polynomial, which didn't follow the fluctuations and 'smoothed' the trend. This is observable in the plots for hourly data in the results section and for other granularities in appendices A-C.

A possible subject of concern constitutes the high degrees of polynomial models: for weekly, daily and hourly data they varied between 50 and 71, creating the threat of overfitting. This tension is relieved by errors that are still at a satisfactory level and are thus high enough to contradict overfitting allegations.

6.2. Complex models

6.2.1. XGBoost

The XGBoost model is characterised as highly efficient, flexible and portable, meaning it can learn complex problems in large datasets. Additionally, it falls under the Gradient Boosting framework and utilises parallel tree boosting to solve complex data-related problems. Hence, XGBoost is well-equipped to deal with the non-stationarity and high granularity characteristics of our data. Gridsearch was employed alongside the model to tune its hyperparameters and find the optimum form of the model to explain the data.

The model was first run with the basic data i.e. weather data and wind production volume as the label. A lagged variable for the label and a rolling mean with a window size of three were included to acknowledge the time-component of the data. Afterwards, two alternative versions were run to address the non-stationarity challenge and explore a market-related explanation behind wind production respectively.

Firstly, a log transformation of wind production was included as a label instead of the normal volume. This transformation decreases the variation in energy production which in turn promotes a more stationary trend. Unfortunately, while the transformation did render the data more stationary, there are still a large number of spikes and drops in the data. The log-label label hence did not produce significantly better results as it consistently majorly mispredicts drops in the data. Please consult [appendix E](#) for modelling results and graphs.

The second variation entails including price data in the form of dynamic electricity prices (excluding taxes) for consumers as a feature in the models. Overall, it improved predictions but not as much as desired, indicating that this specific market indicator is unable to improve model RMSEs performance substantially. [Appendix F](#) contains the modelling results i.e. RMSEs and illustrative graphs on the price models.

As a final overview of the XGBOOST models: three different versions were run for both the onshore and offshore data and for all granularities. This adds to a total of 24 XGBOOST models.

- Only the weather variables (no manipulation of the label)
- Only the weather variables (log-version of the label)
- Weather variables + price (no manipulation of the label)

6.2.2. Prophet

Facebook/Meta's Prophet is frequently applied to business and energy series, combining smooth annual cycles with occasional structural breaks, thanks to its additive decomposition of trend, multiple seasonal components, and optional holiday effects. Prophet's components proved to be a natural fit for our data: the model's changepoints aligned well with the abrupt trend shifts caused by capacity expansions, while its built-in seasonality captured the strong annual cycle - making it a promising choice for our forecasting task.

To stay close to Prophet's native workflow we first collapsed the hourly signal to three resolutions - monthly, weekly and daily means - then built the ds-y frame Prophet expects. We attempted an hourly model, yet even with generous hardware it ran for several hours without converging; we therefore abandoned the experiment and cannot report hourly scores. Nothing else is fed in: the model is purely univariate, so time itself is the single explanatory feature. This decision keeps computation light but inevitably deprives Prophet of wind information. The production volume is passed in without any log or variance-stabilising transformation so that the piecewise linear trend can mirror the physical step-ups in installed capacity that we observed in 2020 and again in 2023.

Parameter choices were intentionally simple. Every model - monthly, weekly and daily - was run with yearly seasonality switched on; the daily variant received an additional weekly seasonal term to absorb work-week fluctuations. We left growth linear, relied on Prophet's default changepoint detector and skipped logistic caps or regressors.

The results are mixed: Prophet captures the broad annual pulse reasonably well at monthly scale but its accuracy collapses at the finer resolutions, where un-modelled weather spikes dominate the variance. In simpler terms, Prophet does a good job handling sudden jumps in capacity and yearly patterns, but it misses the sharp weather-related ups and downs that matter most at weekly and daily levels.

6.2.3. Chronos

Where Prophet tries to explain structure with a handful of transparent components, Chronos, Amazon’s sequence-to-sequence transformer, aims to discover said structure automatically through self-attention. The architecture seems tailor-made for our multi-scale challenge, since attention heads can relate any time step in the input context to any other.

We kept the experiment deliberately minimal. Like Prophet, a pre-trained `chronos-t5-small` checkpoint received only the raw production series (no weather, no prices) and the label was left un-transformed. Token limits restricted the input and output lengths: Chronos can generate at most 64 future time steps, so we excluded daily and hourly resolutions entirely and worked only with monthly and weekly aggregates.

The results are moderate: Chronos performs slightly worse than Prophet and SARIMAX at monthly scale and shows comparable performance on weekly data. Simply put, the model captures some long-range patterns, but its forecasts remain fairly general and do not yet reflect the full complexity of wind production.

Two factors likely explain the gap. First, without fine-tuning, the model’s generic time-series priors do not match the idiosyncrasies of our domain. Second, by withholding wind speed and other meteorological drivers, we denied the transformer access to the very signals that trigger the sharp ramps we want to predict. Nevertheless, once equipped with richer features and a short fine-tune, Chronos could become a powerful way to tackle the non-linear, multi-scale dependencies that are hard to code by hand.

7. Modelling Results

This section evaluates how our six models perform across the four temporal resolutions defined in Section 5.2. Root-mean-square-error (RMSE) is our performance metric of choice to compare the results of each model. Throughout the results section and for each granularity, we contrast the strongest classical baseline with the best complex learner. The offshore and onshore results are compared together in each granularity section as the results are identical i.e. the best performing model for monthly data is the same for onshore and offshore data. The graphs with prediction results of the outperformed models are included in appendices A-D.

7.1. Models' Performance Table

Tables 7.1 and 7.2 below display the RMSEs of our six models for all granularities and both location specifications. For ease of comprehension and comparison, both the baseline and complex model with the lowest RMSE for each granularity is highlighted. This already reveals a significant conclusion from our project, namely that SARIMAX dominates in the baseline league for all granularities on which it was fit i.e., monthly, weekly and daily. The complex models, conversely, paint a more diffuse picture as different models perform better for different temporal resolutions.

We draw two additional key conclusions from this table. Firstly, the offshore and onshore results mirror each other, meaning that the same baseline and complex models perform best for both production locations on the same granularities. This is unsurprising as the structure of the data is similar for offshore and onshore wind production (see graphs 4.3 & 4.4); models that perform well when predicting wind production for a specific granularity, will do so for either location specification.

Secondly, the RMSEs are consistently higher for the onshore compared to the offshore models, meaning all models perform worse when predicting energy production from wind turbines on land. The discussion delves deeper into this phenomenon, for now it suffices to remark that this is likely due to the heterogeneity in onshore production as the location of wind turbines and weather circumstances are much more diffused compared to its offshore counterpart.

Table 7.1: Models' Performance (RMSE) for Offshore Wind Energy Production in the Netherlands

Models' Performance (RMSE) — Offshore Energy Production				
Model	Monthly	Weekly	Daily	Hourly
SARIMAX	279 130	681 348	1 123 375	-
Ensemble	349 879	853 631	1 294 038	1 537 748
Holt-Winters	335 054	856 629	1 724 935	2 275 645
XGBoost	893 069	1 205 043	1 368 290	843 568
Prophet	343 193	861 173	1 252 612	-
Chronos	644 173	696 223	-	-

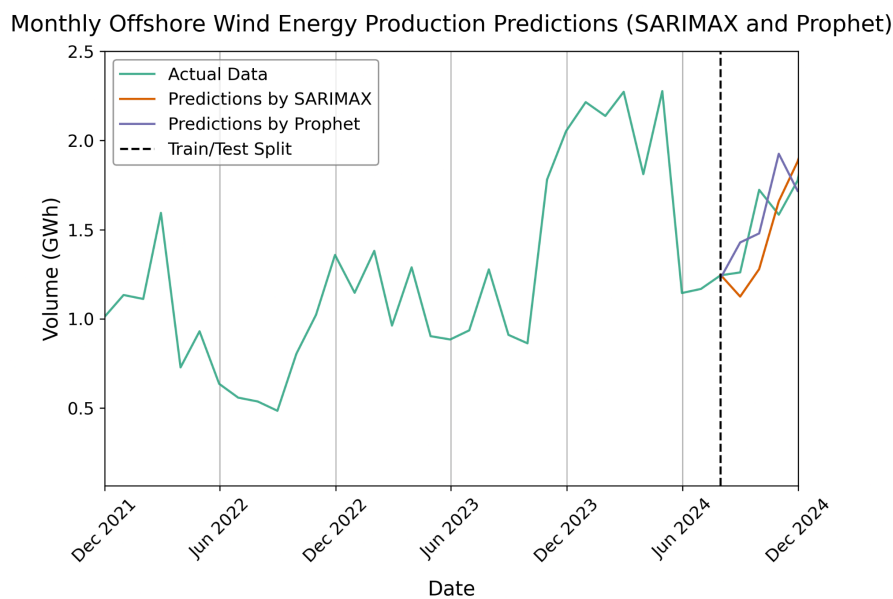
Table 7.2: Models' Performance (RMSE) for Onshore Wind Energy Production in the Netherlands

Models' Performance (RMSE) — Onshore Energy Production				
Model	Monthly	Weekly	Daily	Hourly
SARIMAX	179 815	819 913	1 260 901	-
Ensemble	414 909	971 487	1 461 657	1 625 782
Holt-Winters	331 502	962 509	2 086 193	2 480 649
XGBoost	983 691	1 258 568	1 451 051	891 734
Prophet	383 068	983 646	1 359 270	-
Chronos	469 942	750 766	-	-

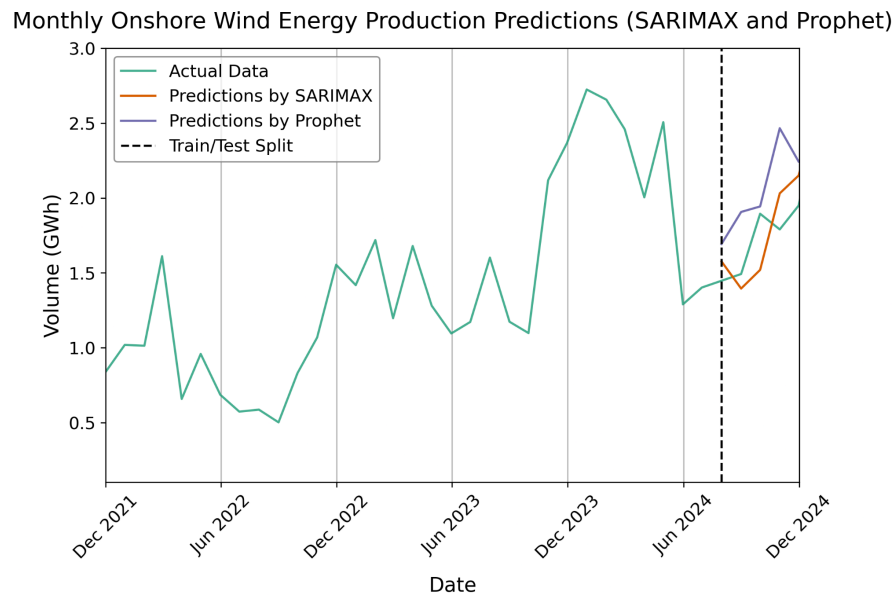
7.2. Monthly data

At a 30-day cadence even violent weather spikes average out, leaving trend and seasonality to explain most variance. Consequently, the SARIMAX specification that embeds a single seasonal lag (12 months) delivers the smallest RMSE for both location specifications. Prophet narrows the gap by explicitly modelling changepoints but cannot quite match SARIMAX because its piece-wise linear trend under-reacts to the 2020 and 2023 capacity jump. The machine-learning contenders lag far behind: XGBoost has trouble with the sparse 96-row training set, while Chronos cannot compress a multi-year context into its limited token budget without fine-tuning.

Graph 7.1 Monthly Offshore Wind Energy Production: SARIMAX and Prophet



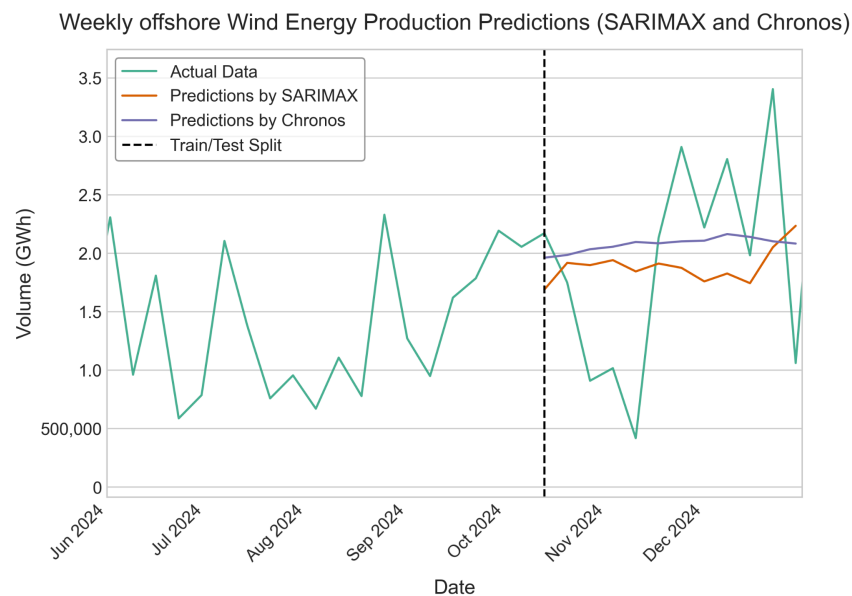
Graph 7.2 Monthly Onshore Wind Energy Production: SARIMAX and Prophet



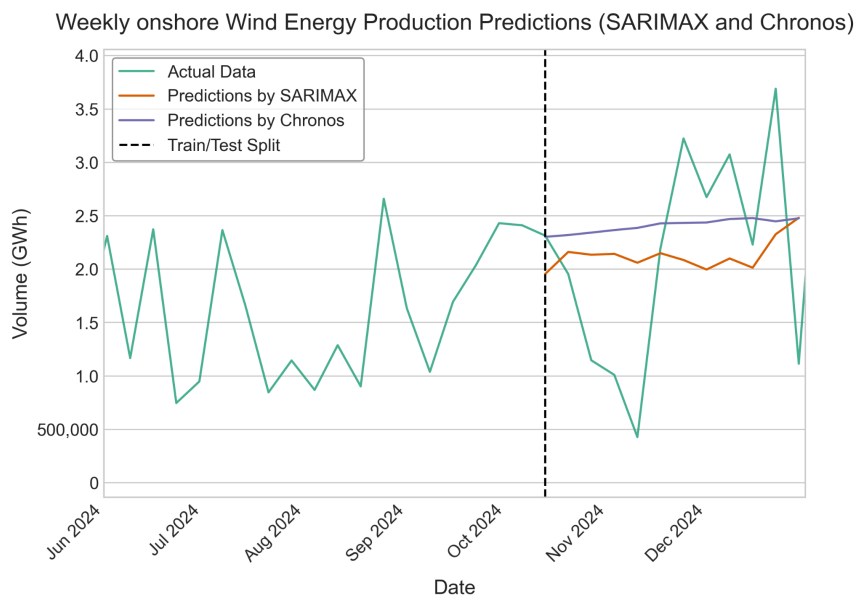
7.3. Weekly data

Weekly aggregation retains some intra-year noise, exposing models to the sharp winter-vs-summer contrast plus short cold fronts. Offshore, SARIMAX again leads, benefitting from its seasonal component (52 weeks) and autoregressive memory. Onshore, however, Chronos edges ahead by ~9 % thanks to its attention heads, which latch onto recurrent wind-direction patterns that escape the linear SARIMAX filter.

Graph 7.3 Weekly Offshore Wind Energy Production: SARIMAX and Chronos



Graph 7.4 Weekly Onshore Wind Energy Production: SARIMAX and Chronos



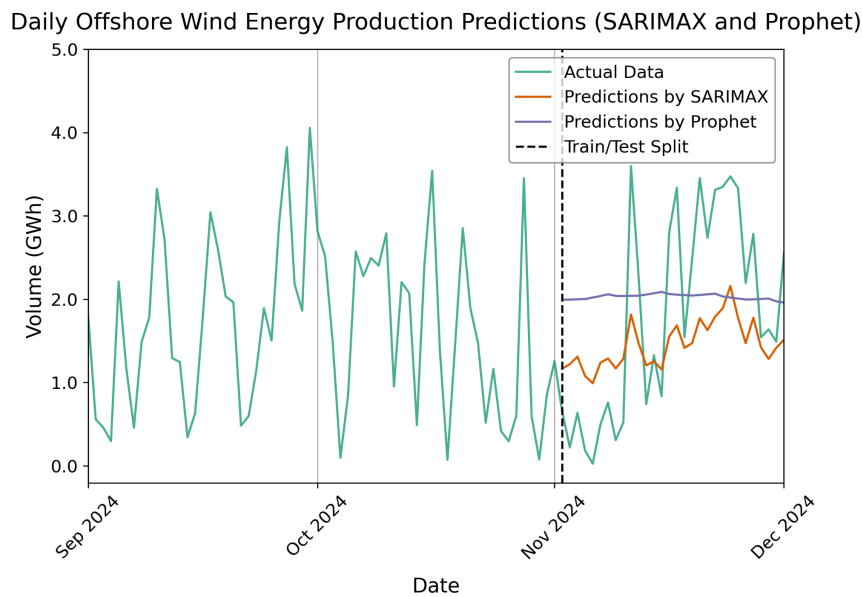
7.4. Daily data

When moving to the daily granularity level, wind energy data exhibits pronounced heteroscedasticity and wind-driven volatility that impact modelling results. SARIMAX continues to serve as our benchmark for both offshore and onshore datasets, though its comparative advantage diminishes at this level.

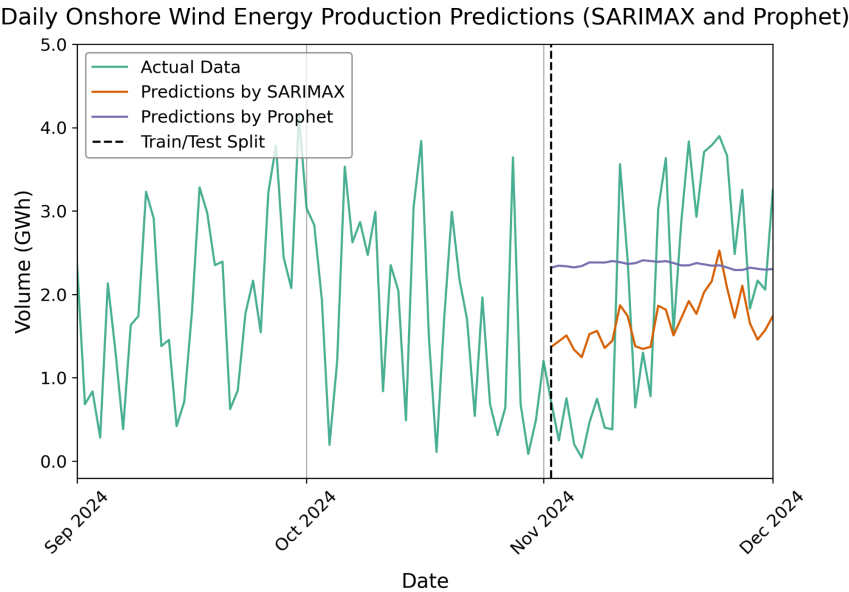
SARIMAX's inability to capture the substantial variability of daily data likely stems from our approach – specifically, the application of seasonal decomposition (STL) to manage computational constraints. By subtracting the annual seasonality component (365-day cycle) from the daily data, we made the modelling computationally feasible. However, this approach also averaged patterns across multiple cycles, smoothing out the inherent variability of daily predictions.

Prophet is effectively blind at this scale without exogenous wind-speed data, so their RMSEs exceed SARIMAX by roughly 10–25 %. This is because by withholding wind speed and other meteorological drivers, we denied the transformer access to the very signals that trigger the sharp ramps we want to predict.

Graph 7.5 Daily Offshore Wind Energy Production: SARIMAX and Prophet



Graph 7.6 Daily Onshore Wind Energy Production: SARIMAX and Prophet



7.5. Hourly data

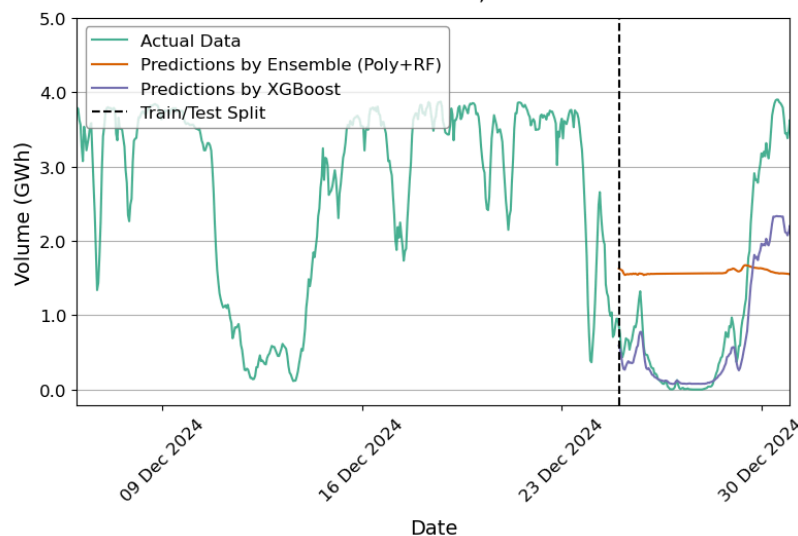
The hourly data was by far the most complicated granularity to model due to the large size and high volatility. Given these complicating factors, only three models were able to fit the data, namely Holt-Winters, Ensemble and XGBoost with the latter two producing the best results. XGBoost slashes offshore RMSE by ~45 % compared with the best ensemble baseline and does likewise for onshore.

It is noticeable that the polynomial-random forest combination that returned the lowest prediction error simply neutralizes the volatility of data and effectively predicts the average value with rare, light and localized fluctuations, thus revealing that it is still trying to grasp the changes in trend. This might have occurred due to the 5% (low) impact of Random Forest (which was responsible for short-term volatility) on final predicted values.

XGBoost performs exceptionally well at the hourly time window compared to the other models and its own predictions as the RMSE for daily and weekly granularity was much higher. The model's strength when confronted with complicated and highly granular data shines here. However, it is noticeable that the model is underpredicting production peaks and likely overfitting as it perfectly mirrors all smaller spikes and drops in the actual data. When looking at feature importance, the rolling mean and lagged variable are heavily relied upon which reaffirms the overfitting conclusion (see graph 7.9). This is further corroborated by looking at XGBoost predictions before the rolling mean and lagged features were included in the model as illustrated by graph 7.10 since the predictions' line is much more erratic and follows the actual data patterns less closely.

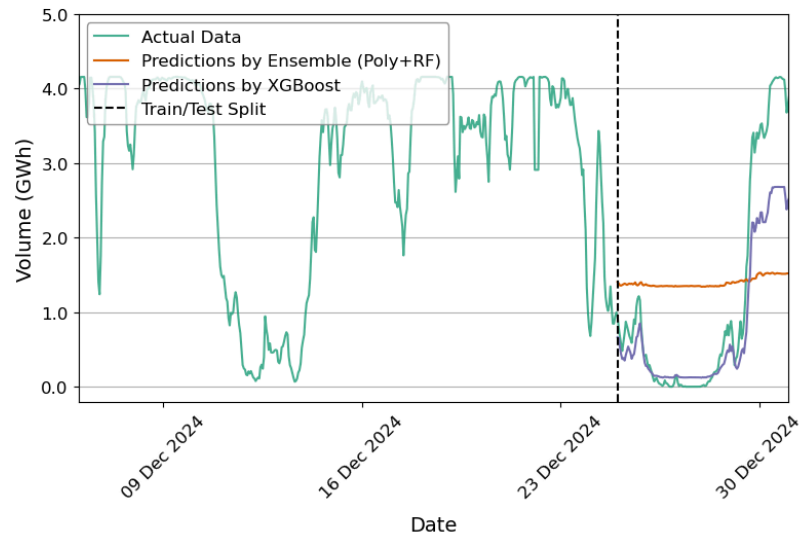
Graph 7.7 Hourly Offshore Wind Energy Production: Polynomial + Random Forest and XBoost

Hourly Offshore Wind Energy Production Predictions (Polynomial+Random Forest and XGBoost)

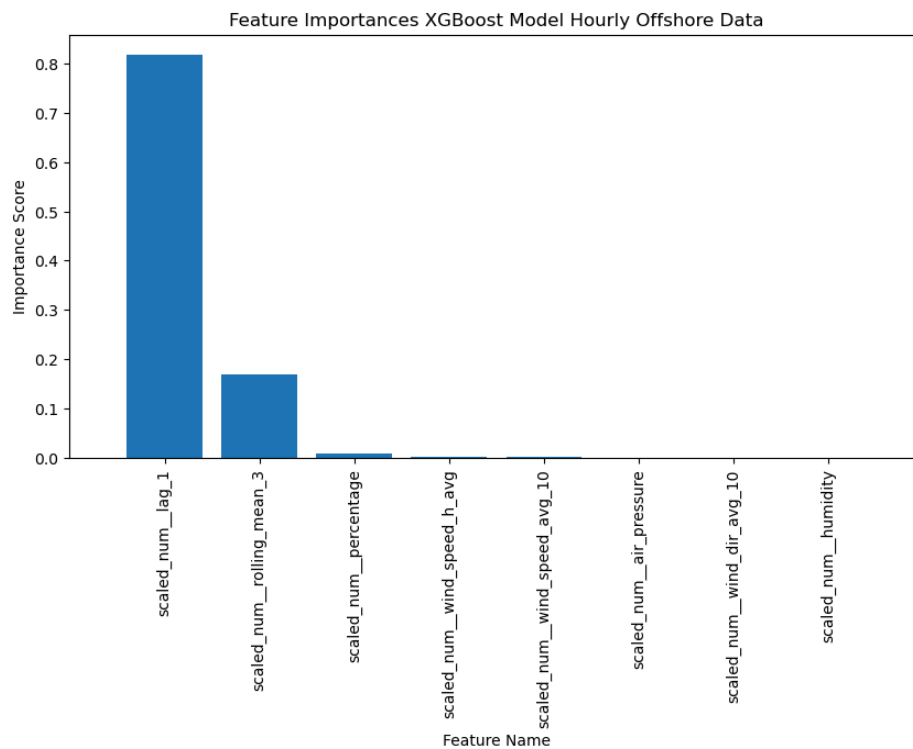


Graph 7.8 Hourly Onshore Wind Energy Production: Polynomial + Random Forest and XBoost

Hourly Onshore Wind Energy Production Predictions (Polynomial+Random Forest and XGBoost)

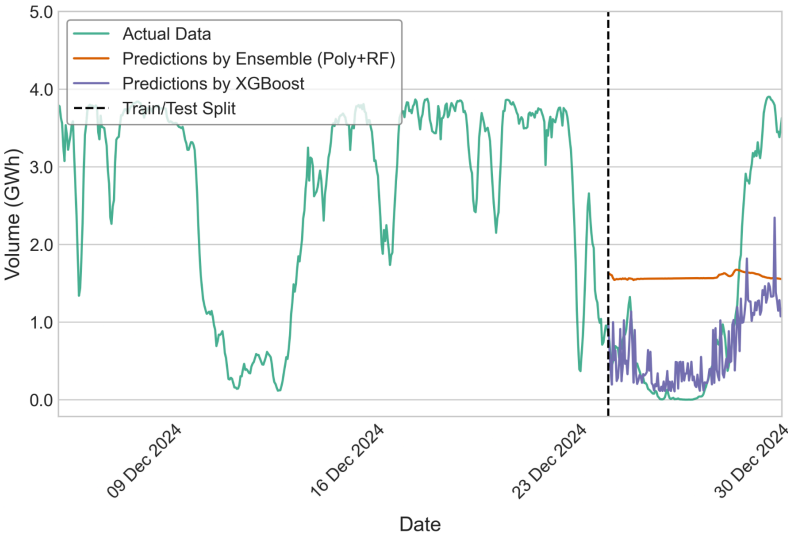


Graph 7.9 Feature Importance XGBoost Model Hourly Offshore Data



Graph 7.10 Hourly Offshore Wind Energy Production: Polynomial + Random Forest and XBoost without rolling mean or lagged variable

Hourly Offshore Wind Energy Production Predictions (Polynomial+Random Forest and XGBoost)



8. Discussion and next steps

In this section, we interpret these results and present two factors that likely negatively impacted our ability to predict the data alongside possible solutions to improve our current approach. We conclude the discussion by highlighting two avenues to expand the project's scope.

8.1. Interpretation

Our results for monthly data follow the overall upward trend but do not capture the final spike. The monthly data is characterised by very few data points and comparatively few fluctuations, thus embodying a trade-off between simpler data and less training points. Both the baseline and complex model capture the upsloping trend but have a difficult time fully grasping the data. Despite the incompleteness, the models' overall good performance is reflected in the comparatively low RMSEs. The monthly granularity is therefore useful as an initial point for prediction but not informative to predict wind energy for policy or economic purposes.

Regarding the weekly and daily granularity, our predictions form a smooth line instead of capturing the upwards and downwards trends revealed by the actual wind production. These flat prediction lines reveal weaknesses in our model specifications: Sarimax subtracted the annual seasonality component to enable fitting the data, whilst Prophet could not include exogenous features. Subsequently, the models are unable to capture smaller spikes and trends in the data. Therefore, our model selection is able to fit weekly and daily granularities but not accurately enough to inform improved grid management.

Regarding the final granularity, hourly data, we conclude that the high granularity and non-stationarity made it exceptionally hard to model, which is why only three models were able to fit the data. The ensemble model results in predictions that form a smooth, almost straight line, just like Chronos and Prophet for the weekly and daily data. This is because, for high granularities, ensemble models with a weighting balance skewed towards the polynomial model leave little room for the random forest's contribution to short-term fluctuations, and such imbalance led to the lowest errors.

XGBoost, on the other hand, is the only model that follows spikes and drops in highly granular data. This is because the model is especially adequate when confronted with data like ours. A major caveat to these impressive results is the fact that XGBoost is regularly both overfitting and underpredicting. The overfitting likely results from the heavy reliance on the features rolling mean and lag-label to predict future values.

Additionally, models like XGBoost are apt at discovering and understanding complex patterns in complex data, yet encounter difficulties when tasked with extending these patterns beyond the scope of the training data. This is a clear case where XGBoost has understood the structure of the data well but is unable to fully comprehend and subsequently predict future values as it cannot reach the high peaks at the end of 2024. Hence, the predictions for hourly granularities seem most fit for informing grid operators but the risk of overfitting questions future reliability.

To summarise, we successfully addressed the challenge of high granularity by fitting our models to various time windows. This aided our understanding of which models perform best for which granularities and enabled us to explore the strengths of different models. Our second challenge, non-stationarity, was addressed but not fully overcome as our weekly and daily models struggle to capture the various trends and patterns in our data. Although XGBoost does well at capturing these difficult patterns in the hourly data, it relies heavily on lagged and rolling mean features and tends to overfit.

8.2. Complicating Factors

We now move on to discuss two points that likely negatively impacted our modelling results and include recommendations to improve our methods.

1) Spatial aggregation dilutes heterogeneous weather signals.

The Nationaal Energie Dashboard only provides wind production data based on the broad categories ‘offshore’ and ‘onshore’, these are aggregates of all turbines in either space. This ignores the fact that these wind turbines are spatially very diffused, especially the onshore ones. Yet, data on the location and number of wind turbines in the Netherlands is private company information and therefore undisclosed to the public. We could hence not account for the wind turbines’ location, wide spatial disparity nor the number of turbines employed.

Consequently, we resorted to aggregating meteorological features for both offshore and onshore data. Unfortunately, wind regimes differ markedly between and within the North-Sea clusters, the Wadden Islands and inland regions. These complex dynamics were not aided by the fact that we had to manually assign weather stations to the offshore and onshore category before aggregating the data.

It is possible that we did not allocate the weather stations properly meaning that the meteorological data is even less well-tailored to the location specification of onshore vs offshore.

The broad categorisation of onshore and offshore production combined with aggregate meteorological data complicated our prediction task: Spatial differences were smoothed over and the local weather factors that drive hour-to-hour production are not tailored sufficiently. We suspect that this significantly impacted the explanatory strength of the meteorological features and inhibited our ability to predict off- and onshore wind generation with greater accuracy. This would also partially explain why our predictions for onshore data are consistently worse compared to their offshore counterparts for all granularities. Since the diversity in onshore regions and hence weather circumstances is greater, the aggregated data is noisier and offers less help to predict generation.

Solution: To resolve these aggregation-related issues, we require data that includes the location and number of windmills. This would allow us to divide the production data over smaller regions and predict wind energy generation at the local level. We could then also introduce the wind conditions belonging to different geographical locations and hence circumvent the issue of aggregating all onshore and offshore meteorological data.

2) The limited impact of dynamic consumer prices

The XGBoost model was employed to explore whether market indicators aid in the prediction of wind generation. The rationale behind this addition was that market dynamics in terms of supply, demand and price are in constant communication with each other. Hence, dynamic prices which depend on the contemporary supply and demand of electricity were considered as a potential indicator of the electricity produced. However, while the inclusion of a price indicator did improve XGBoost's RMSEs, model performance did not see a major improvement (see [Appendix F](#)).

This lacking effect is likely due to the electricity market's complex nature. This market includes different prices for different time windows (futures, day-ahead and intra-day) and a diverse set of actors. Furthermore, wind energy is often sold in long-term contracts and enjoyed considerable state support through Power Purchasing Agreements (PPA's). Subsequently, its production has fewer implications for consumer consumption and dynamic prices than other types of electricity generation like solar power. It is therefore overly simplistic to consider our price feature as a fitting representation of all market dynamics. Yet, due to time and data availability constraints, this was the most accessible indicator.

Solution: Given more time and experience with the ENTSO-E API, we could improve our range of market indicators by including intra-day, day-ahead and future prices. These short- and long-term prices are likely far better indicators for wind production as they directly affect wind producers' decision to increase or downsize generation.

8.3. Future Research

To conclude the discussion section, we recommend the following possibilities to go beyond the scope of this project, namely explore more complex models and expand the geographic scope to other countries.

1) Explore hybrid spatio-temporal ensembles

High-capacity architectures like Chronos-T5 can, in theory, capture complex multi-scale patterns—but only if they are fed sufficient and well-structured data. In our case, the transformer was trained with very few aggregated observations (weekly data includes a mere 418 observations) and no domain-specific fine-tuning. Subsequently, the model trailed behind more data-efficient models like SARIMAX for the weekly and monthly time specification.

This underperformance is not only attributable to the lack of data volume: deep networks such as LSTMs and ConvLSTMs expect higher-order input structures - 3D tensors for sequences, 5D arrays for spatio-temporal dependencies. The publicly available wind datasets that we utilised do not offer that level of detail and we were hence not able to train these complex models to exploit their full strength.

As a next step, future researchers could try to access or collect richer and more spatially disaggregated data that include the necessary dimensions, and then use this enhanced data to finetune models like Chronos or LSTM. This requires building a pipeline for high-order feature tensors - 3D inputs for LSTMs and 5D arrays (batch $\times$ time $\times$ lat $\times$ lon $\times$ channel) for ConvLSTM - populated with down-scaled weather fields and categorical land or sea marks.

2) Expand the geographic scope

An interesting method to test the validity of our models and approach would be to test their ability to predict wind generation in other countries. Viable case studies include countries with similar production dynamics like Denmark (both onshore and offshore generation) and countries without offshore generation like Switzerland. The former's context should translate well considering the similarities in weather and generation patterns. The latter, however, has a vastly different terrain, energy market and grid system. These comparisons would illuminate how well our models and approaches are able to adapt to similar and new circumstances and so if other countries can employ our research to aid their grid strategy.

9. Conclusion

This project set out to develop and evaluate machine learning models for forecasting offshore and onshore wind energy production in the Netherlands, addressing two core challenges inherent in such a task: the non-stationarity of time series data and the exceptionally high temporal granularity of electricity generation data. Through a carefully structured multi-resolution modelling strategy, we assessed the performance of a diverse range of models—from basic machine learning approaches to advanced time series analysis—across both offshore and onshore wind energy data.

One of the clearest outcomes of our analysis is the strong performance of the SARIMAX model across monthly, weekly, and daily resolutions for both off- and onshore energy production. Despite its relative simplicity compared to complex learners like Prophet or Chronos, SARIMAX's ability to model seasonality, trends, and autoregressive components made it a highly effective tool for structured and moderately granular time series. This reinforces a broader lesson in machine learning: model complexity should be tailored to data characteristics, and in many real-world scenarios, simpler models, when correctly specified, can outperform more sophisticated ones.

However, when moving to hourly granularity—the most challenging resolution in our study—the limits of statistical models became apparent. Here, XGBoost stood out, outperforming all other models by a substantial margin. Its ability to handle non-linear relationships and interact with engineered features like rolling means and lagged values enabled it to approximate the high-frequency volatility characteristic of wind power generation. Yet, this performance came with caveats: signs of overfitting and a tendency to underpredict extreme peaks indicate that even powerful machine learning models struggle to generalize in the presence of sparse, noisy, and volatile patterns.

Our findings also highlight critical limitations of the collected data. The lack of spatial disaggregation in both the production and meteorological variables likely weakened the explanatory power of weather variables, especially for onshore wind turbines, which are subject to greater geographic and climatic variability. Moreover, while dynamic consumer electricity prices added some predictive value, their marginal impact underscores the complexity of the electricity market and the limited influence of end-consumer prices on wind generation, which is often pre-contracted and heavily regulated.

Looking ahead, several avenues for future research emerge. First, improving spatial resolution—both in terms of turbine location and weather conditions—would likely enhance model performance. Second, expanding the model architecture to include spatio-temporal structures such as ConvLSTM could help capture complex dependencies currently beyond the reach of simpler methods.

Third, applying and comparing our modelling framework to other countries with different wind generation and grid characteristics could reveal the robustness and transferability of our approach.

In conclusion, our project contributes a holistic, data-driven exploration of wind energy forecasting using machine learning. While no single model offers a complete solution, our findings demonstrate how a careful alignment of data, model, and granularity can yield meaningful predictions. Ultimately, improving the accuracy of wind power forecasts is essential not only for grid stability and economic efficiency but also for enabling the broader energy transition required to meet climate goals.

Statement of Contributions

Benedict:

- Introduction
- Energy Production Forecasting as ML problem
- Conclusion
- Literature research/Models Research
- Attempt at LSTM
- GitHub

Sofiya:

- Retrieve weather data from KNMI
- Ensemble Model + model description
- Co-author hourly results
- template for graphs final results

Polina:

- EDA
- Chronos + model description
- Prophet + model description
- Co-author monthly/weekly/daily results
- Wind station map website

Kirill:

- Literature research
- Analysis of time series data challenges
- Sarimax + model description
- Holt-Winters + model description
- Co-author monthly/weekly/daily results

Franka:

- Retrieve energy data from NED
- Retrieve price data
- XGBoost + model description
- Co-author hourly results
- Discussion
- Appendix E + F

Everyone:

- graphs with model results own models appendices
- final presentation

Literature

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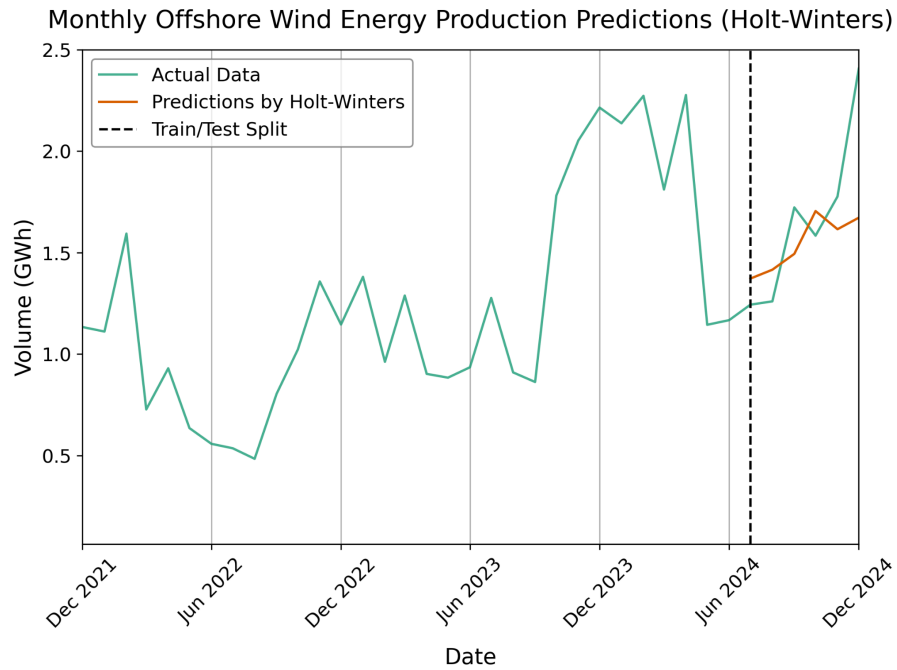
Appendix A: Monthly predictions

SARIMAX

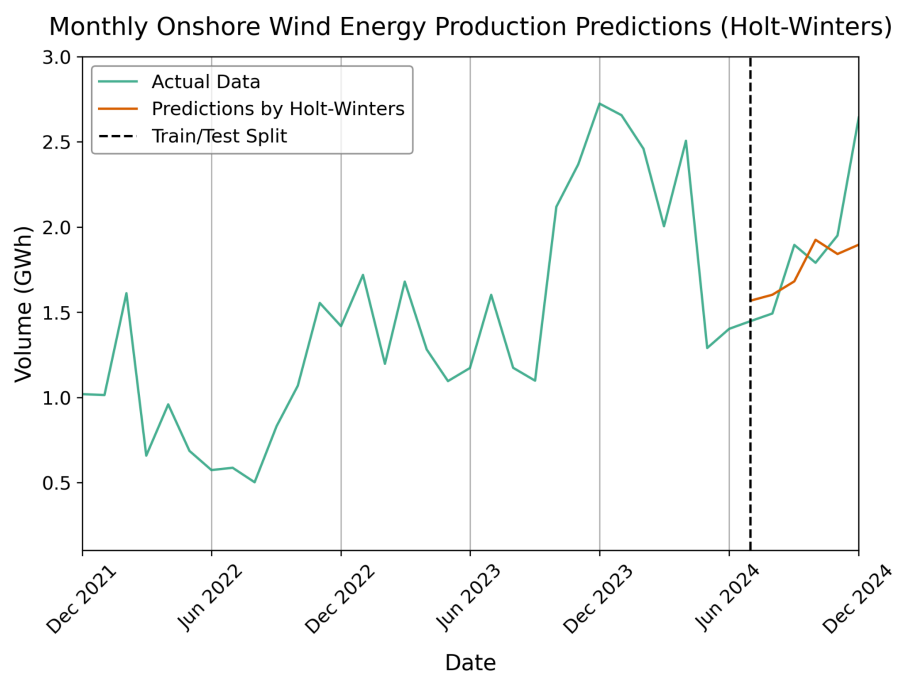
Predictions by SARIMAX models are not included in the Appendix because they are already presented in the Modelling Results section of the Report.

Holt-Winters

Graph A.1 Monthly Offshore Wind Energy Production: Holt-Winters

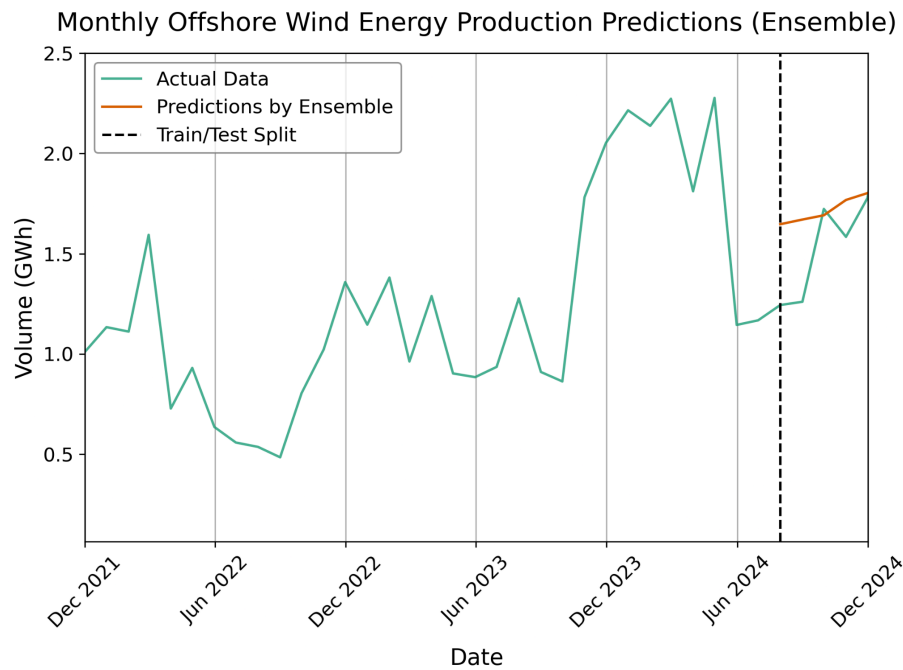


Graph A.2 Monthly Onshore Wind Energy Production: Holt-Winters

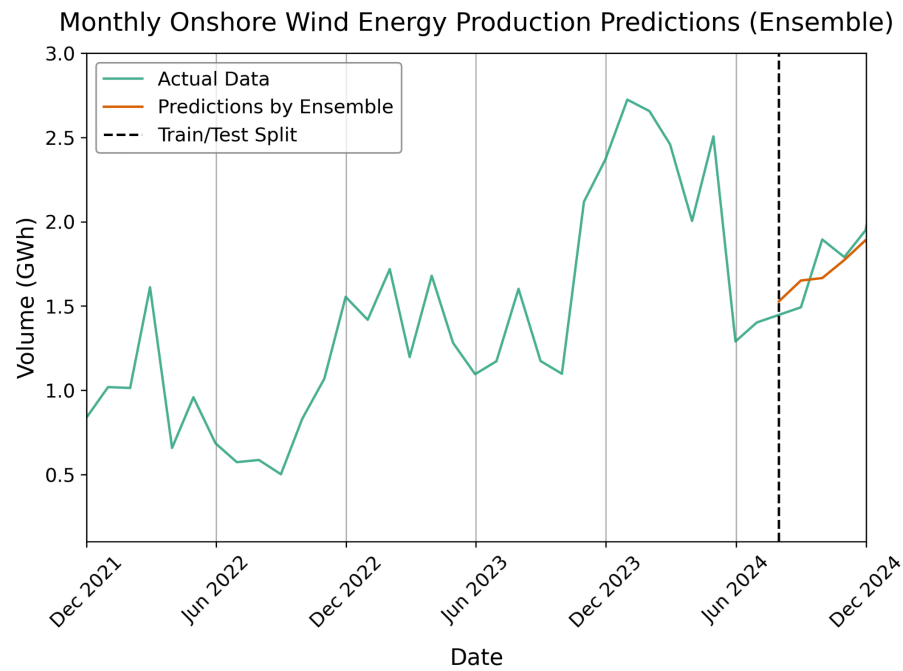


Ensemble

Graph A.3 Monthly Offshore Wind Energy Production: Ensemble

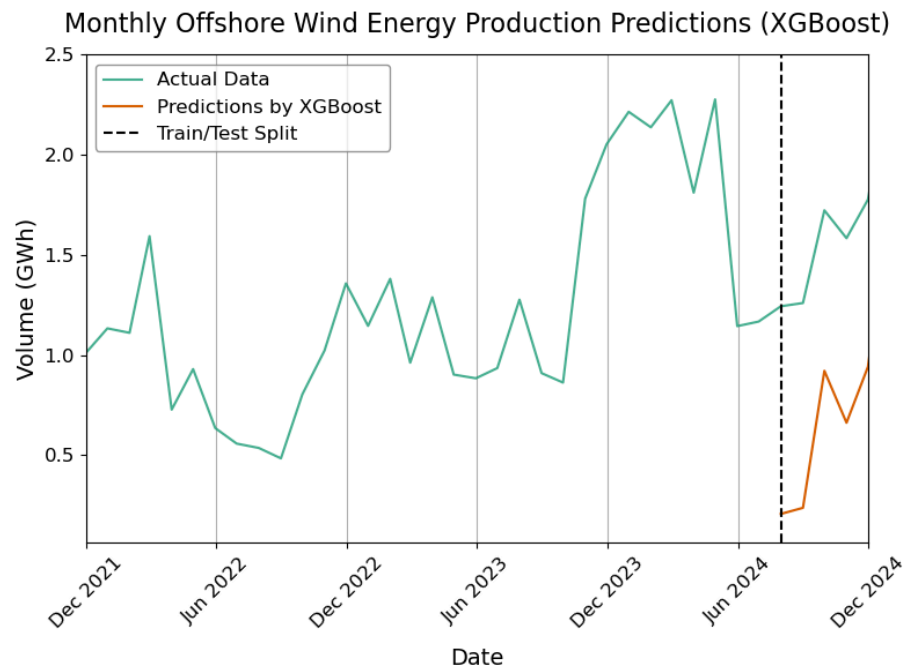


Graph A.4 Monthly Onshore Wind Energy Production: Ensemble

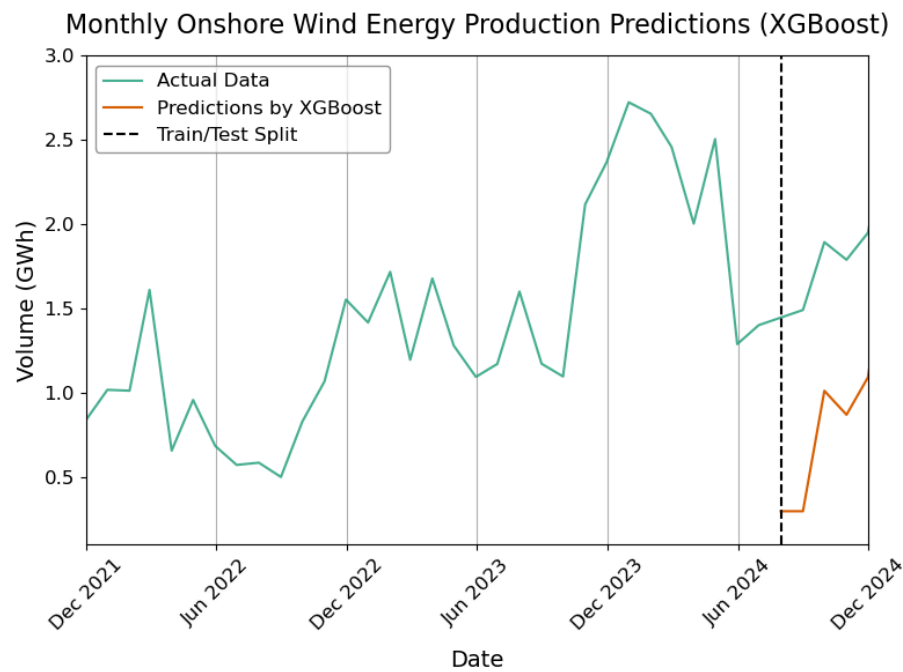


XGBoost

Graph A.5 Monthly Offshore Wind Energy Production: XGBoost

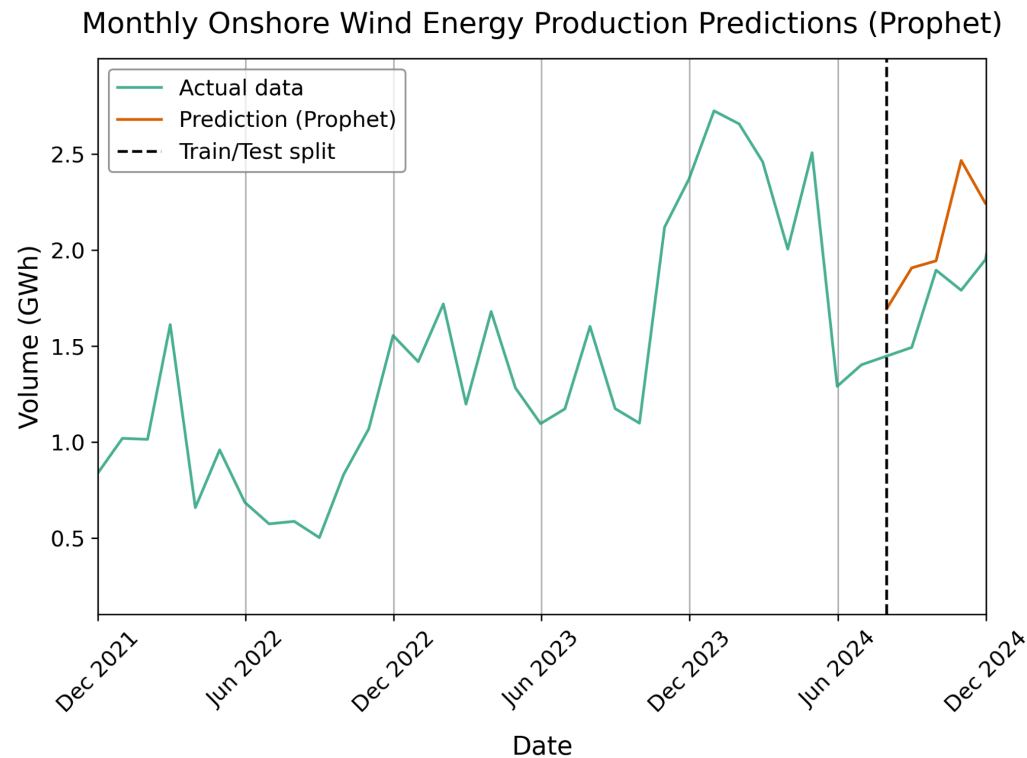


Graph A.6 Monthly Onshore Wind Energy Production: XGBoost

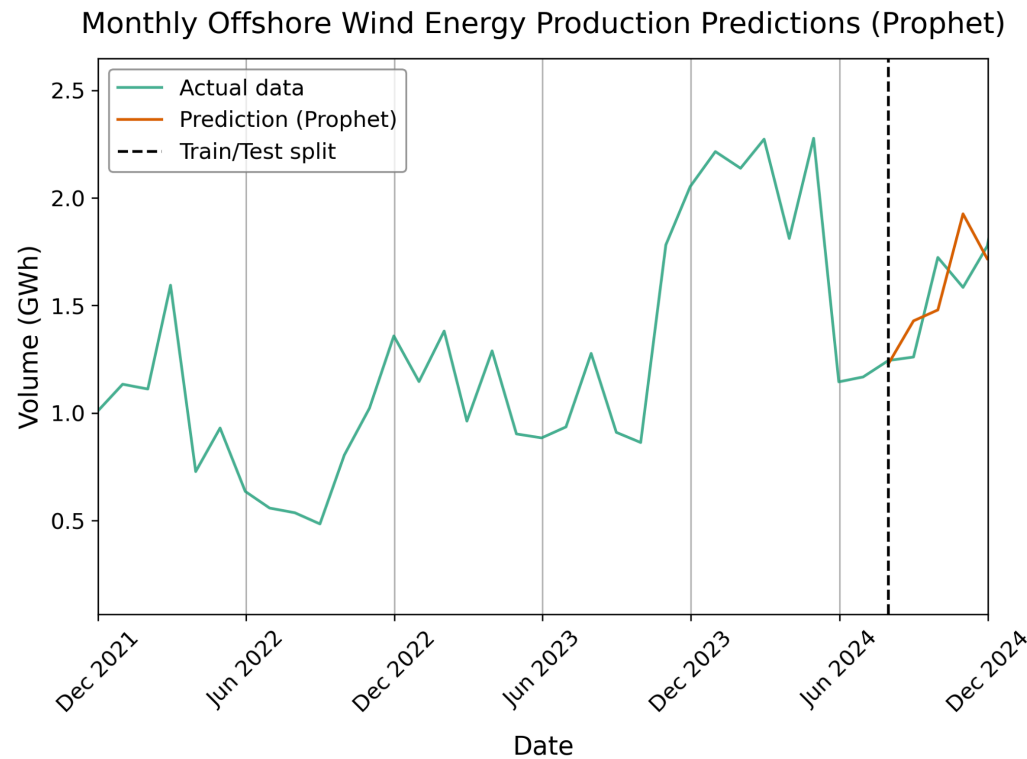


Prophet

Graph A.7 Monthly Onshore Wind Energy Production: Prophet

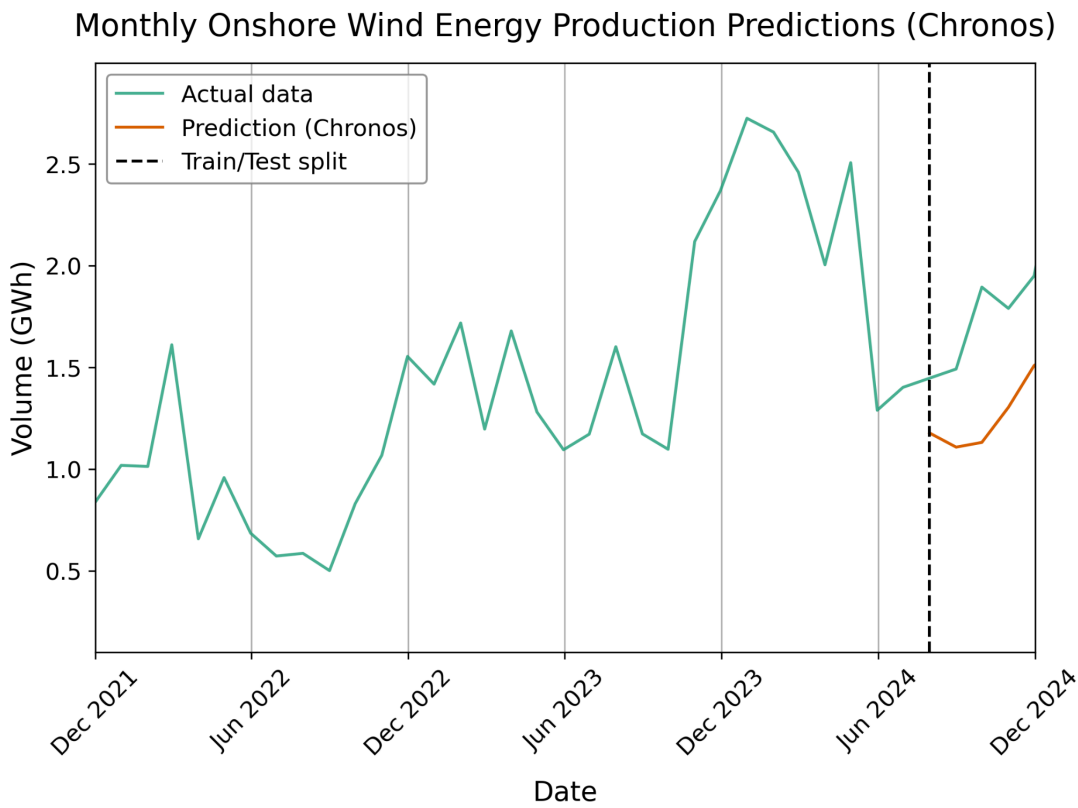


Graph A.8 Monthly Offshore Wind Energy Production: Prophet

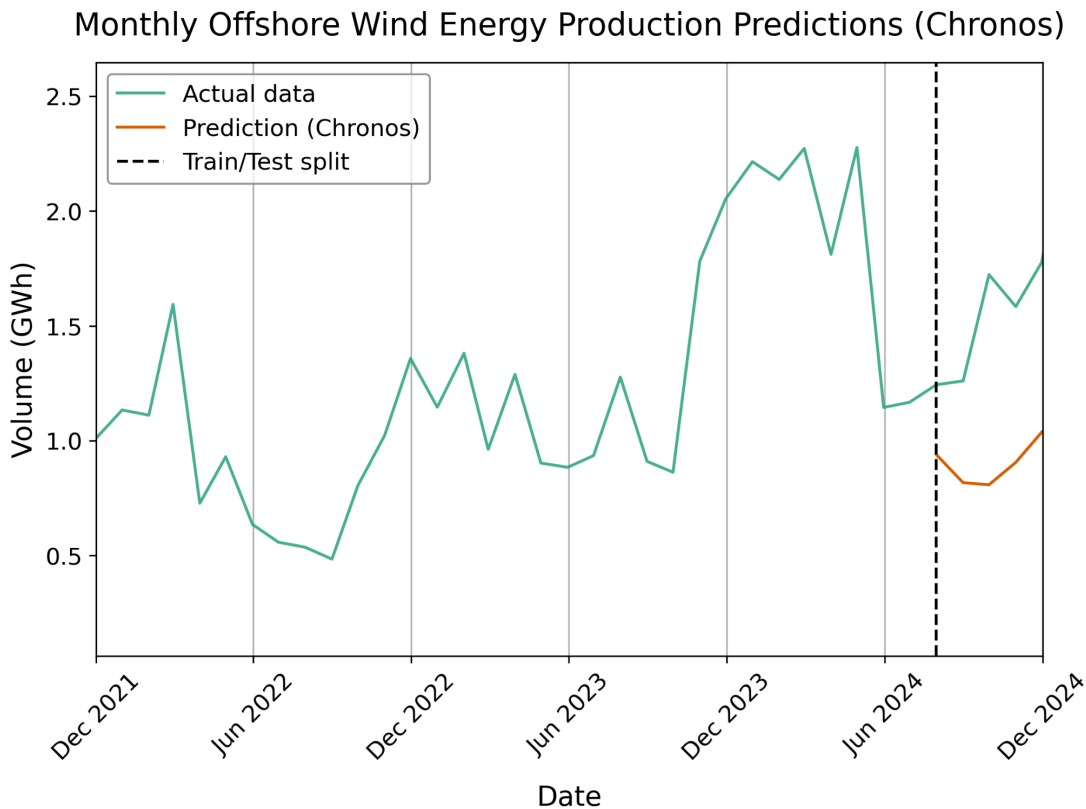


Chronos

Graph A.9 Monthly Onshore Wind Energy Production: Chronos



Graph A.10 Monthly Offshore Wind Energy Production: Chronos



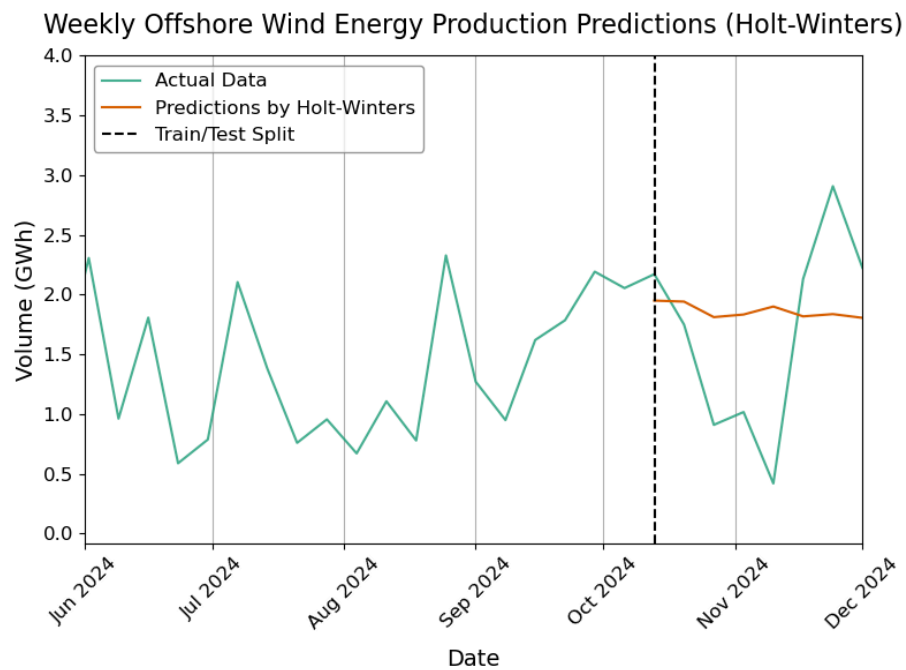
Appendix B: Weekly predictions

SARIMAX

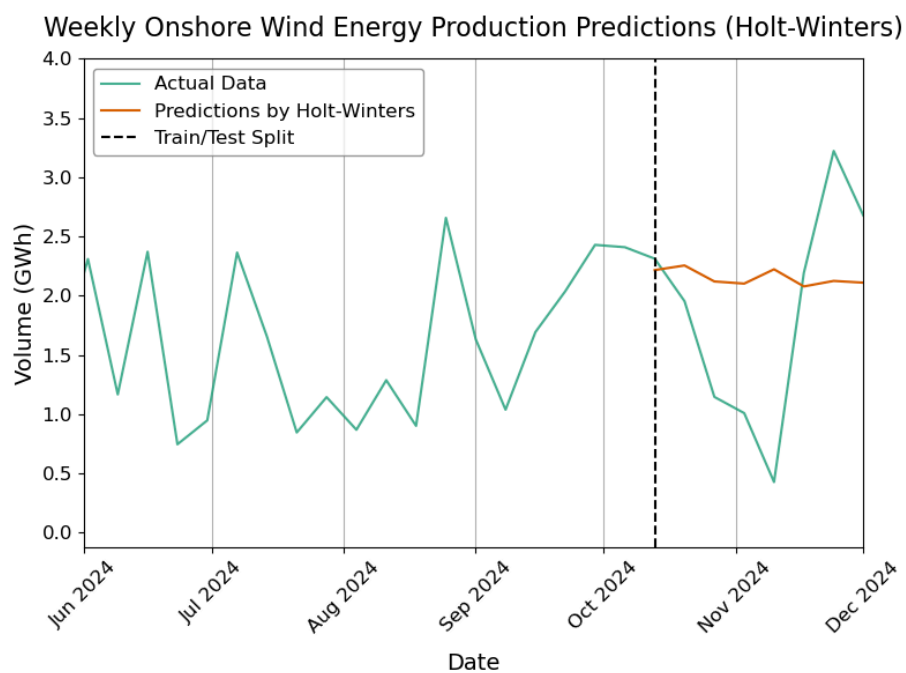
Predictions by SARIMAX models are not included in the Appendix because they are already presented in the Modelling Results section of the Report.

Holt-Winters

Graph B.1 Weekly Offshore Wind Energy Production: Holt-Winters

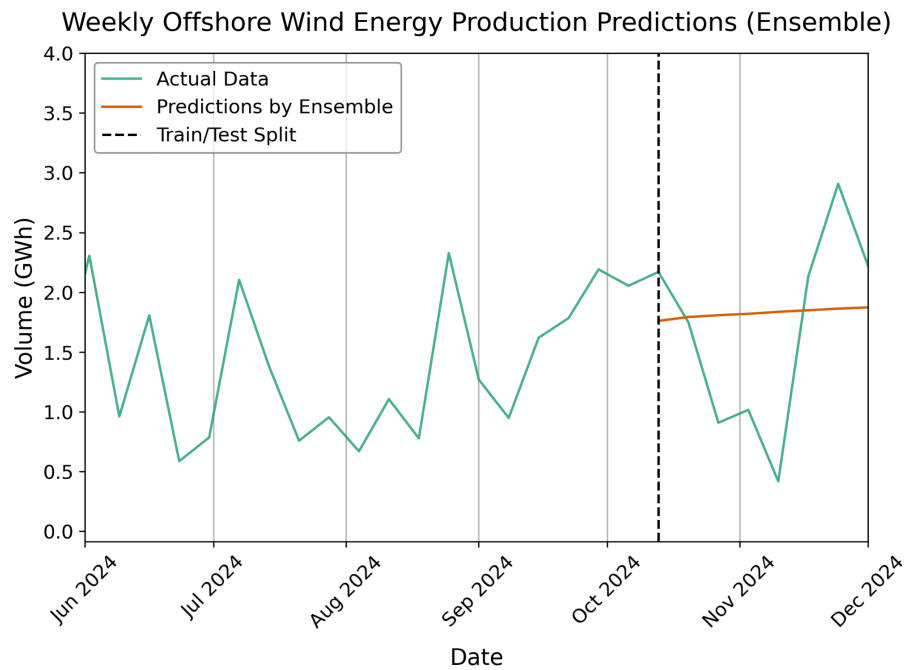


Graph B.2 Weekly Onshore Wind Energy Production: Holt-Winters

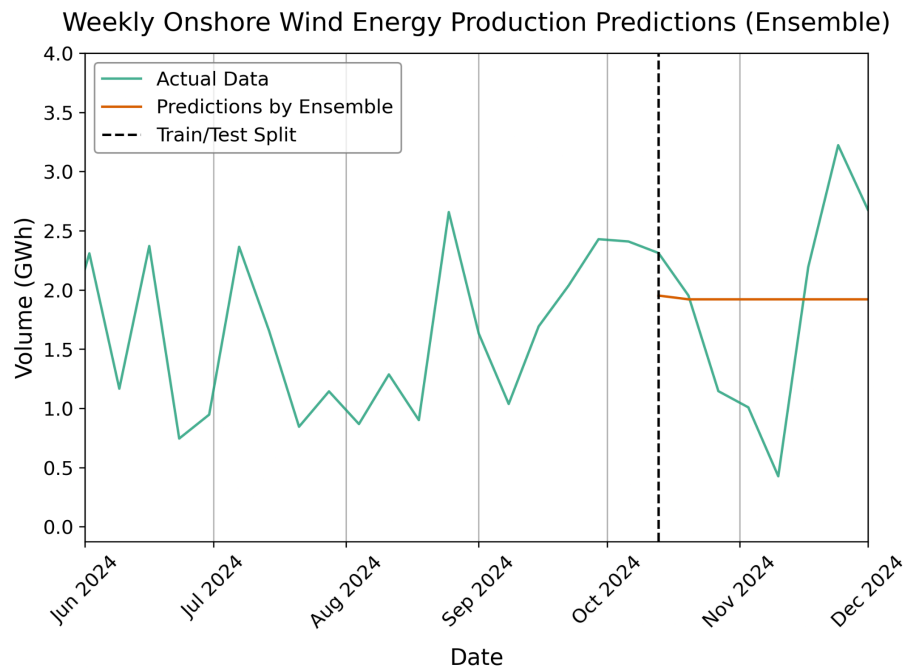


Ensemble

Graph B.3 Weekly Offshore Wind Energy Production: Ensemble

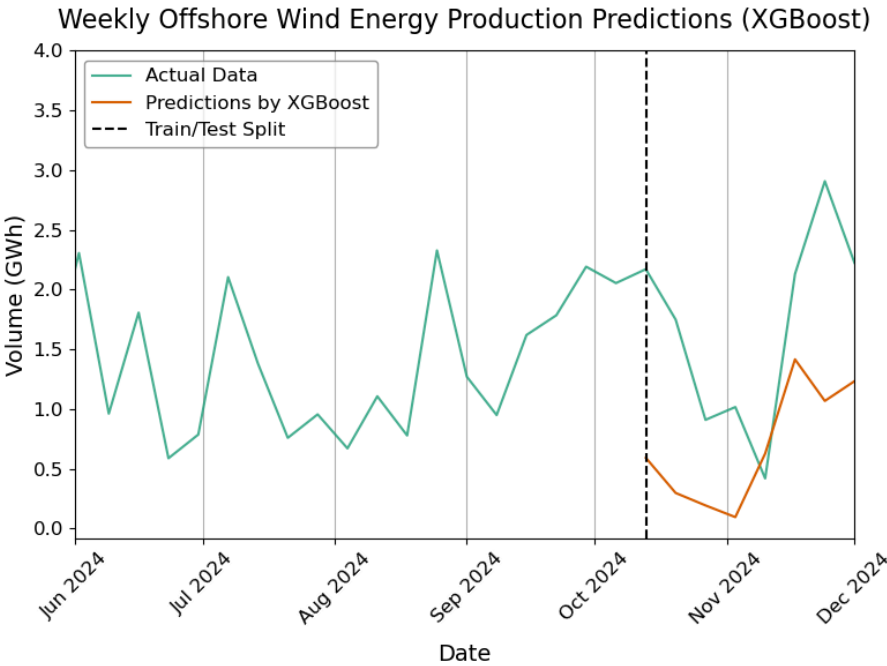


Graph B.4 Weekly Onshore Wind Energy Production: Ensemble

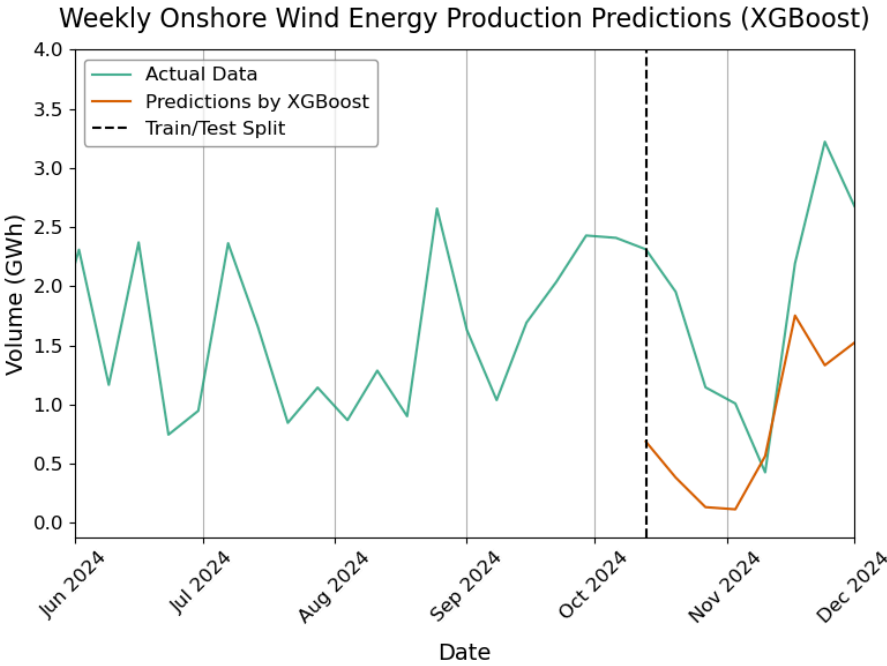


XGBoost

Graph B.5 Weekly Offshore Wind Energy Production: Holt-Winters

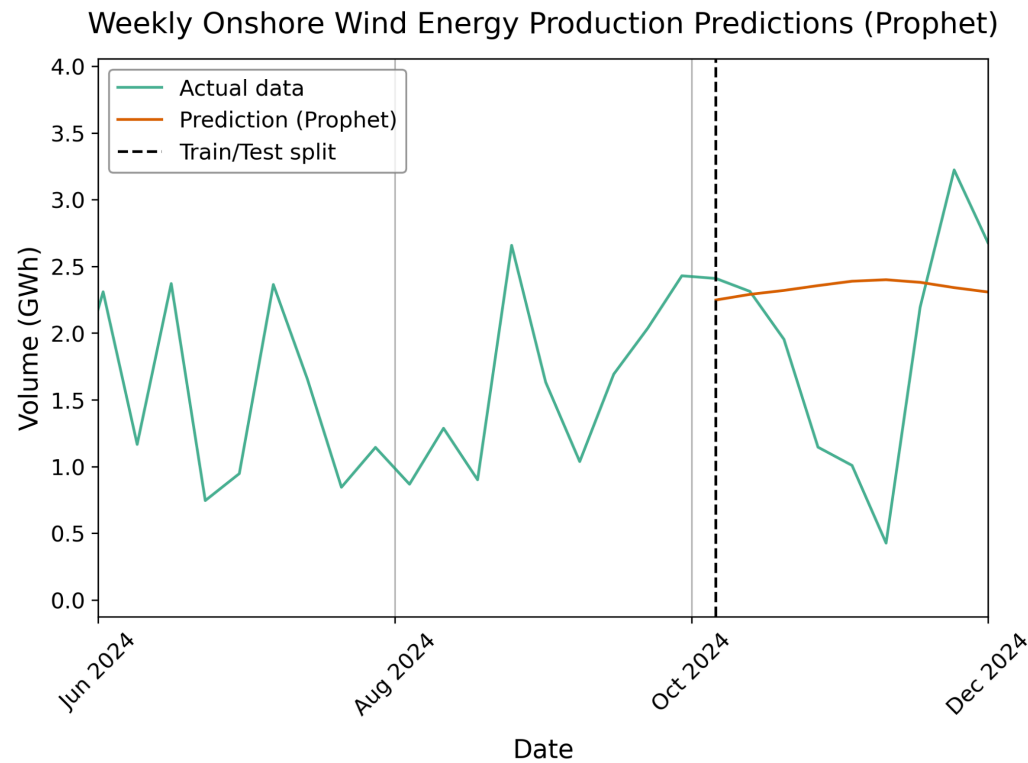


Graph B.6 Weekly Onshore Wind Energy Production: Holt-Winters

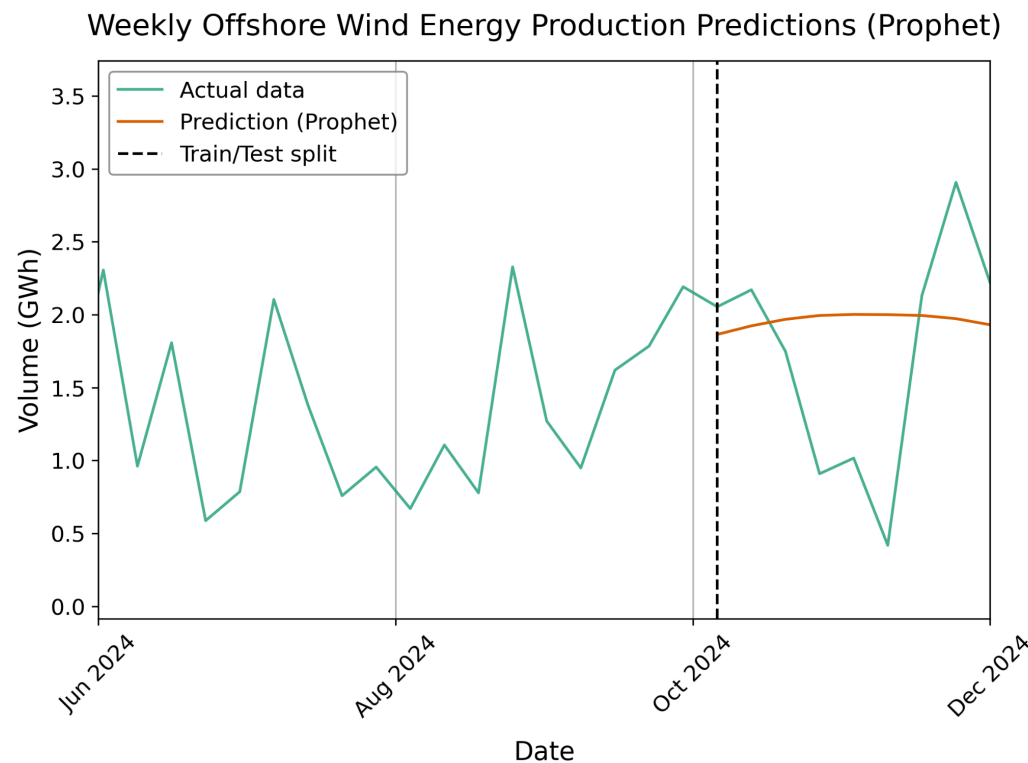


Prophet

Graph B.7 Weekly Onshore Wind Energy Production: Prophet

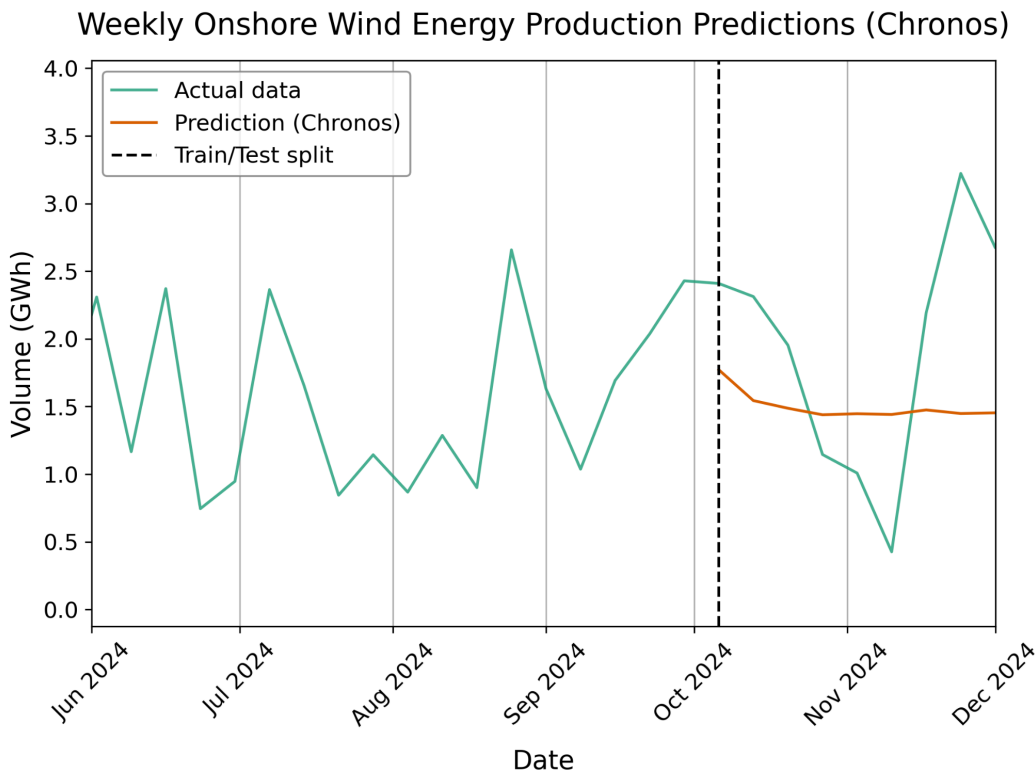


Graph B.8 Weekly Offshore Wind Energy Production: Prophet

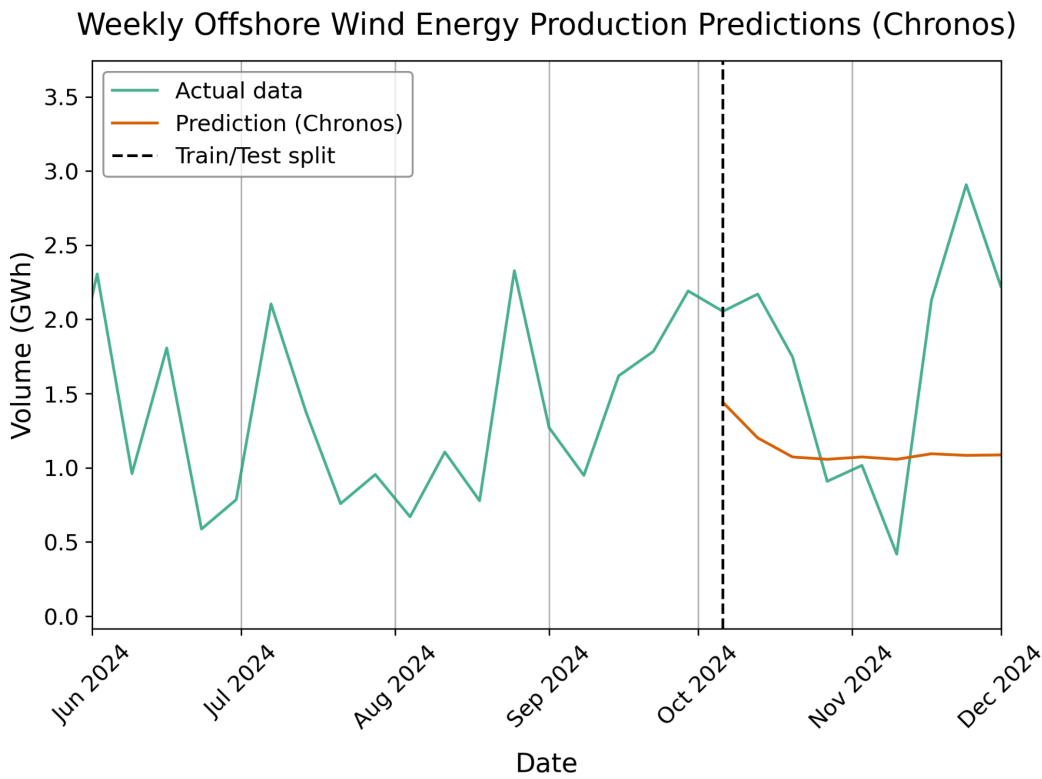


Chronos

Graph B.9 Weekly Onshore Wind Energy Production: Chronos



Graph B.10 Weekly Offshore Wind Energy Production: Chronos



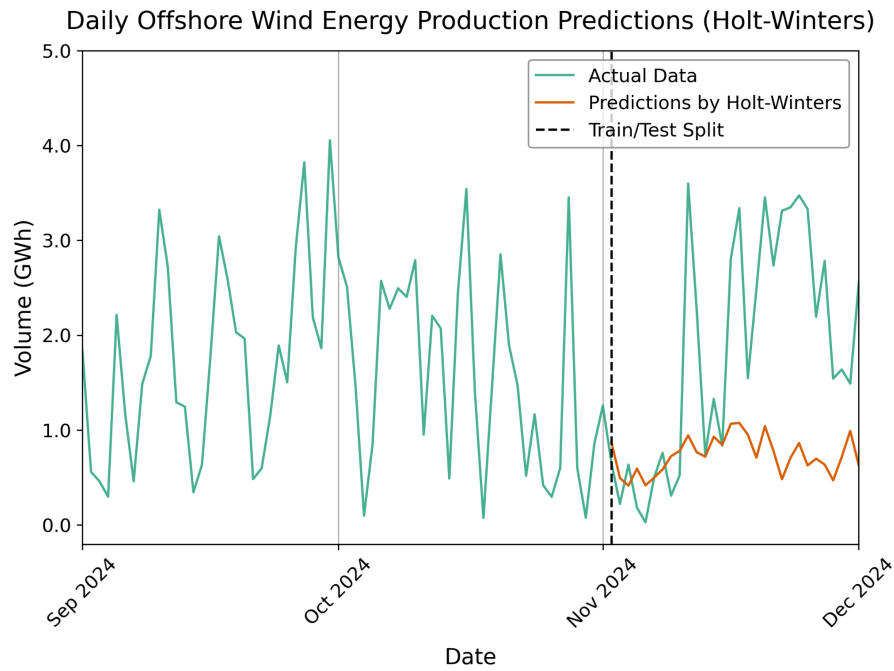
Appendix C: Daily predictions

SARIMAX

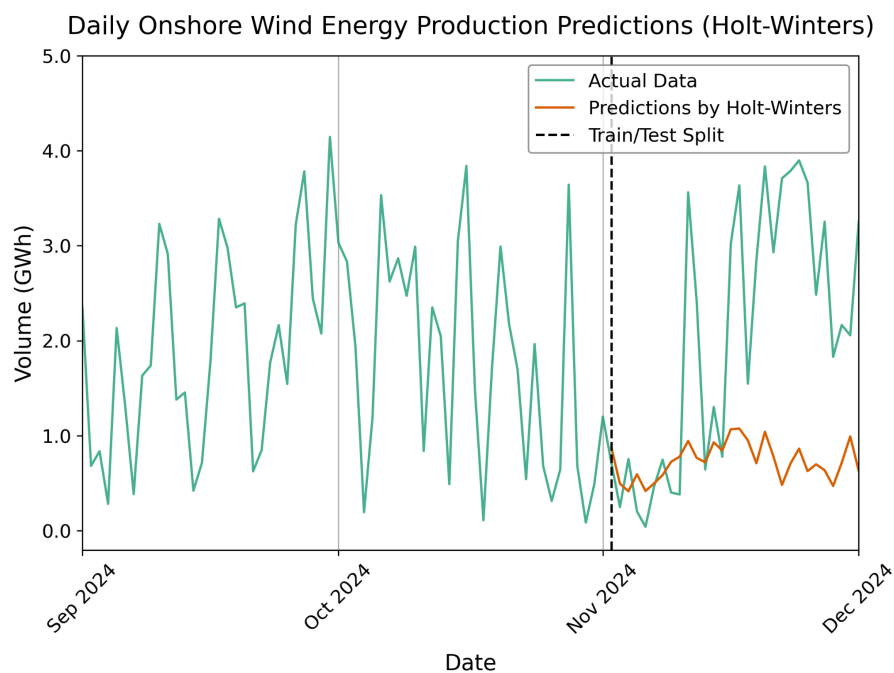
Predictions by SARIMAX models are not included in the Appendix because they are already presented in the Modelling Results section of the Report.

Holt-Winters

Graph C.1 Daily Offshore Wind Energy Production: Holt-Winters

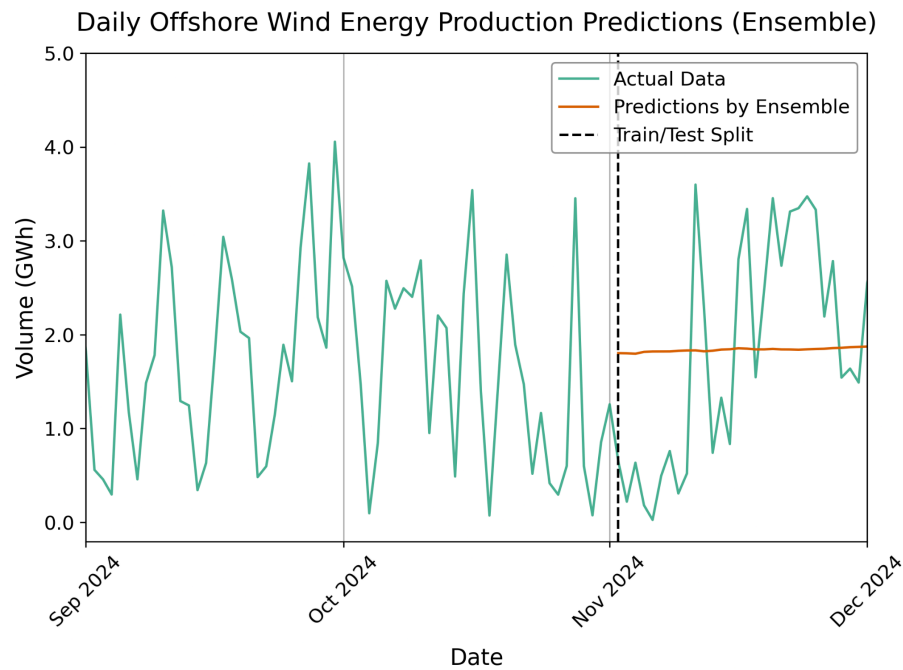


Graph C.2 Daily Onshore Wind Energy Production: Holt-Winters

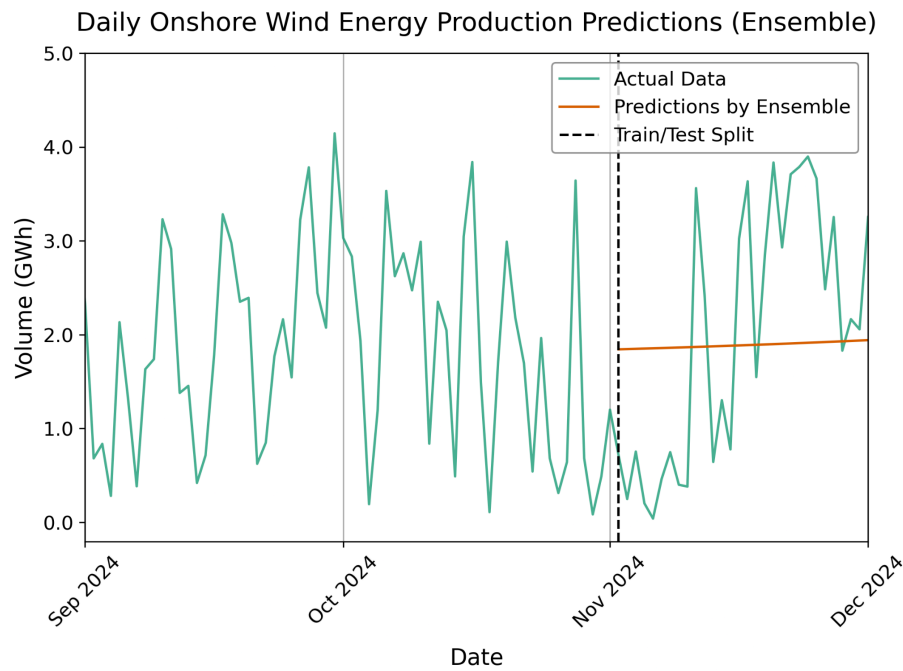


Ensemble

Graph C.3 Daily Offshore Wind Energy Production: Ensemble

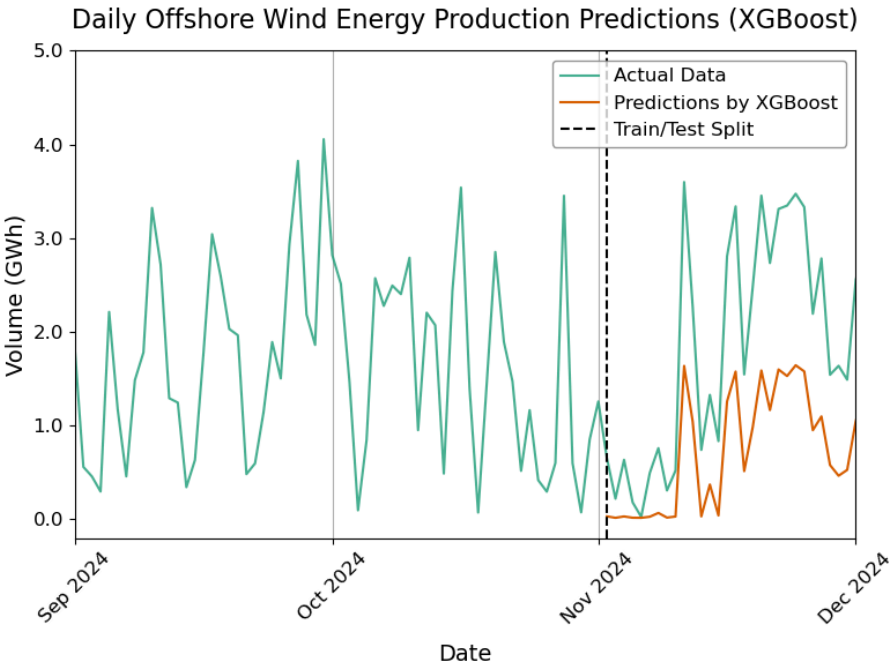


Graph C.4 Daily Onshore Wind Energy Production: Ensemble

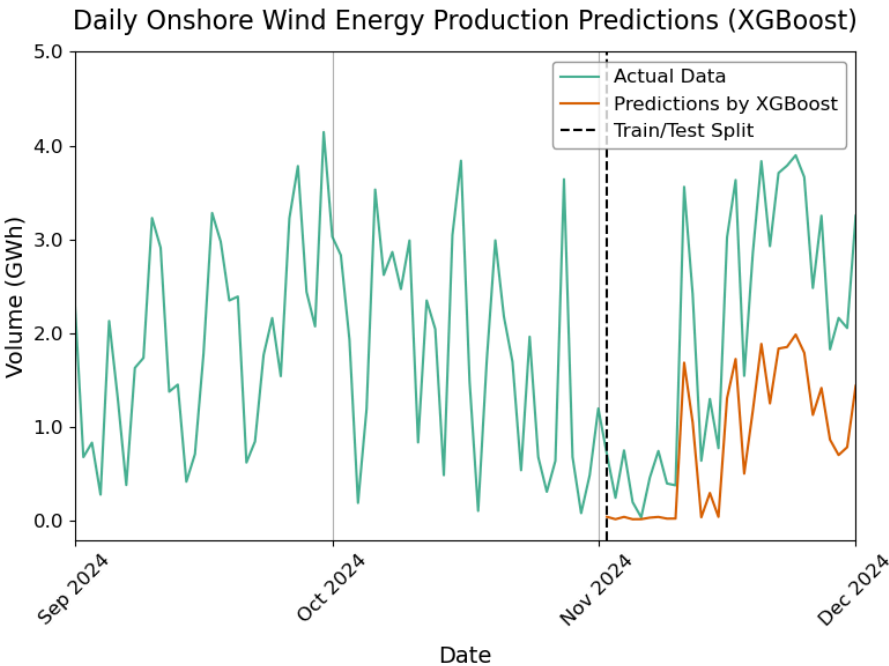


XGBoost

Graph C.5 Daily Offshore Wind Energy Production: XGBoost

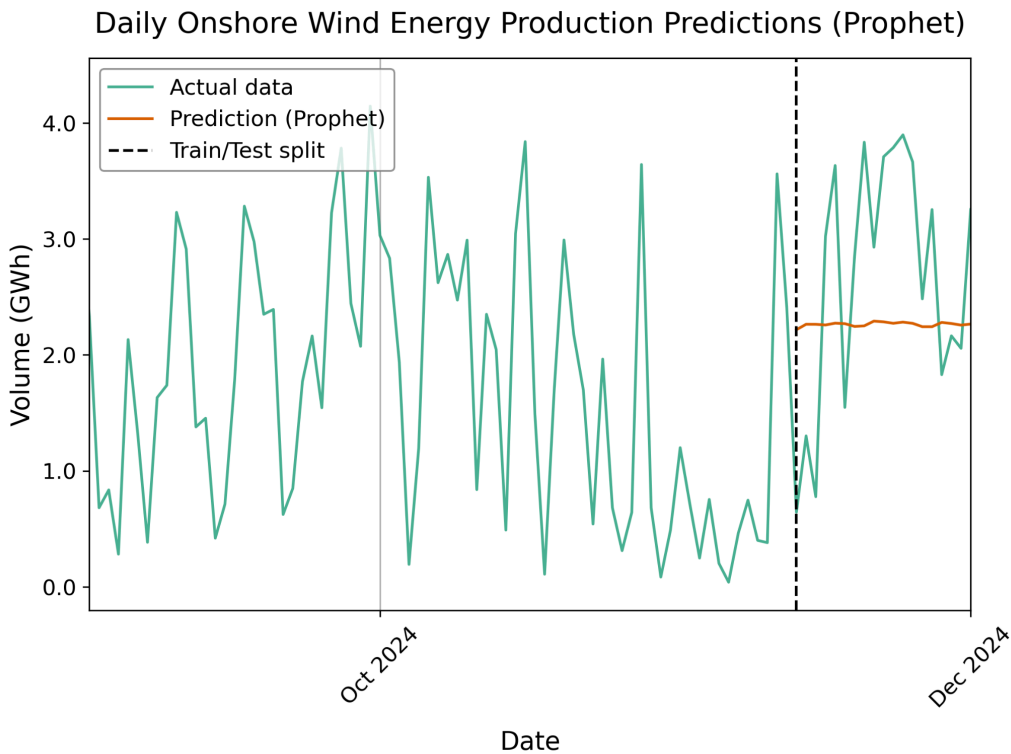


Graph C.6 Daily Onshore Wind Energy Production: XGBoost

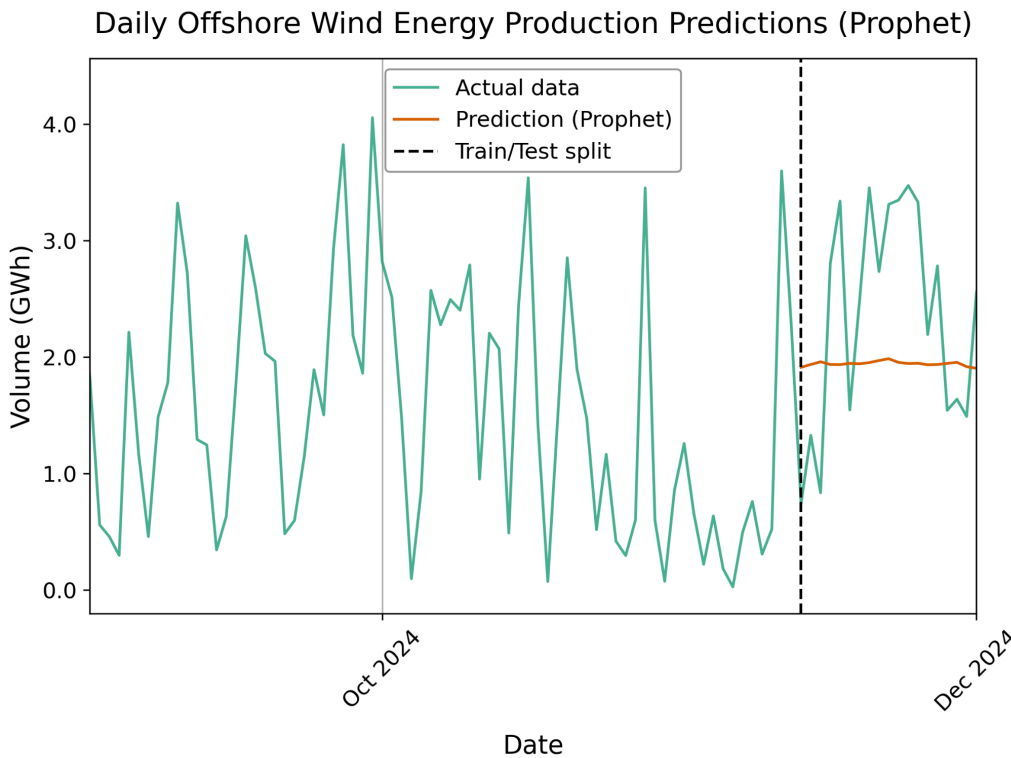


Prophet

Graph C.7 Daily Onshore Wind Energy Production: Prophet



Graph C.8 Daily Offshore Wind Energy Production: Prophet



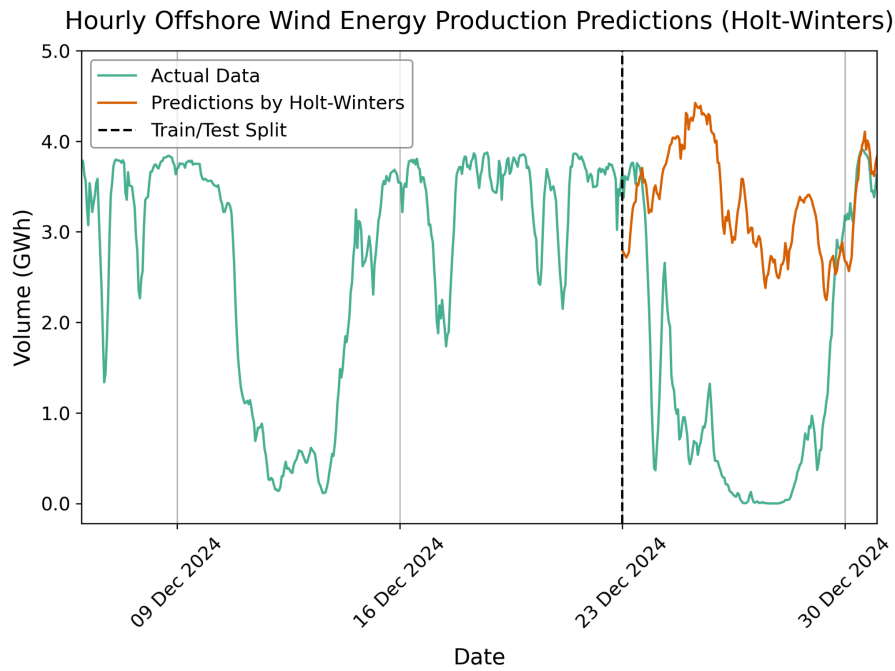
Appendix D: Hourly predictions

SARIMAX

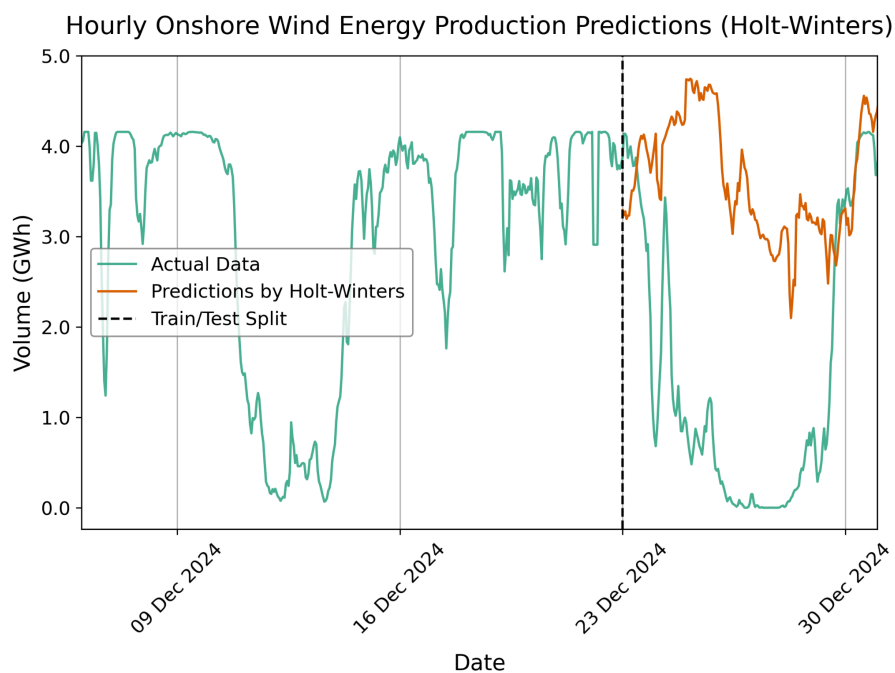
The SARIMAX model is unable to generate predictions for hourly data in reasonable time due to exceptionally high granularity of data (8760 h / cycle).

Holt-Winters

Graph D.1 Hourly Offshore Wind Energy Production: Holt-Winters



Graph D.2 Hourly Onshore Wind Energy Production: Holt-Winters



Appendix E: Log Label Results

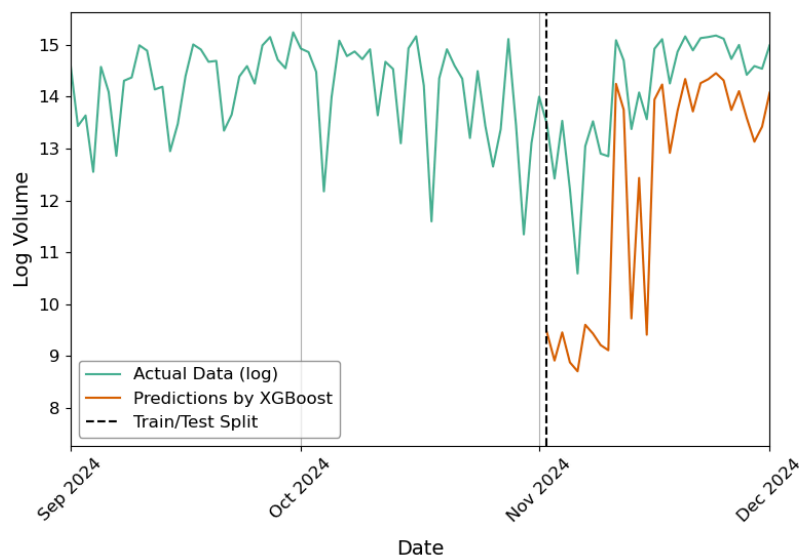
A log-transformation was applied to the production volume: a tried and tested method when working with non-stationary data. And indeed, the data overall became more stationary, compare for instance Graph 15.1 and 15.2, yet still display major spikes and drops. Unfortunately, our predictions do not improve majorly after introducing the log-transformation; XGBoost predicts far greater drops than actually occur in the data. An important deviation from the ‘normal’ label is that offshore is not consistently better predicted here: the MSEs for monthly and hourly granularity are lower for onshore models. Overall, the log-transformation benefitted the data’s stationarity, yet other tools like differencing are perhaps more fruitful options to address the data’s challenges and improve model performance.

Table E.1 XGBoost MSEs for models with Log-Transformation on Volume as Label

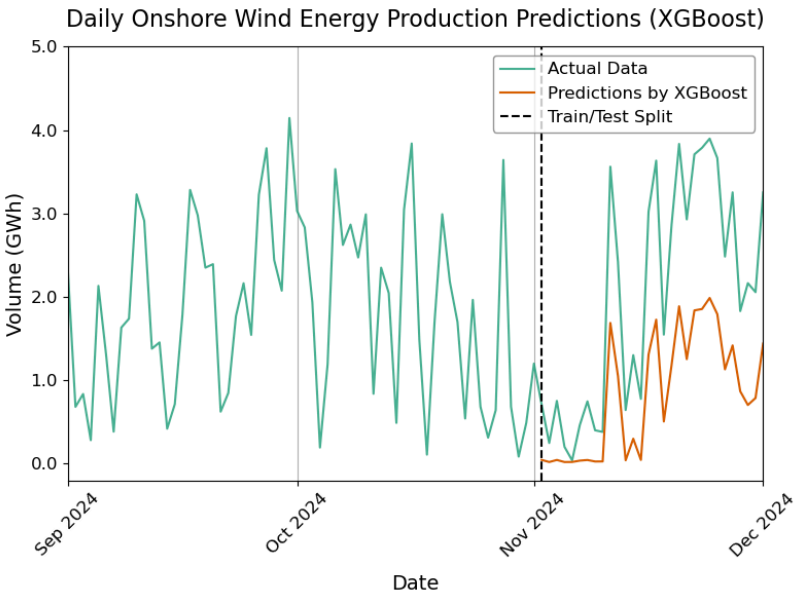
XGBoost Performance (MSE) — On- and Offshore Energy Production with Log Label				
Model	Monthly	Weekly	Daily	Hourly
Offshore	1,138	1,008	3,372	1,657
Onshore	0,948	1,132	3,784	1,141

Graph E.1 Daily Onshore Wind Energy Production: XGBoost with log label

Daily Onshore Wind Energy Production (volume in log) Predictions (XGBoost)



Graph E.2 Daily Onshore Wind Energy Production: XGBoost (no log transformation)



Appendix F: Price Data Results

The inclusion of a market indicator through dynamic consumer prices did not significantly improve XGBoost model performance. Upon inspection of table 16.1, we conclude that the RMSEs are consistently lower but only marginally. Graphs 16.2 and 16.3 concur with our analysis as the prediction lines for XGBoost with and without price data are nearly identical. Nevertheless, when inspecting feature importance for the hourly model, XGBoost did rely on the price feature to explain the data.

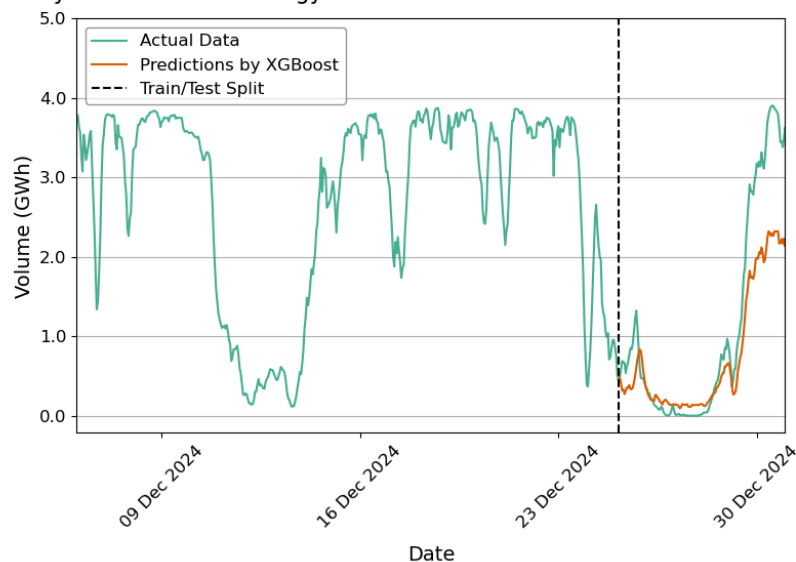
These results support our reasoning behind market dynamics influencing generation and hence being a relevant predictor. However, they primarily indicate the need for more relevant price data as wind generation is largely sold in bulk and not directly to end-consumers. The reason behind the massive improvement for the monthly time window is unclear as the feature importance for the monthly model indicated that the price feature did not contribute, ending in eighth place.

Table F.1 XGBoost RMSEs for models with additional price feature

XGBoost Performance (RMSE) — On- and Offshore Energy Production with Price Data				
Model	Monthly	Weekly	Daily	Hourly
Offshore	471 767	1 204 865	1 364 956	833 400
Offshore (no price data)	893 069	1 205 043	1 368 290	843 568
Onshore	127 567	1 248 250	1 441 700	879 696
Onshore (no price data)	983 691	1 258 568	1 451 051	891 734

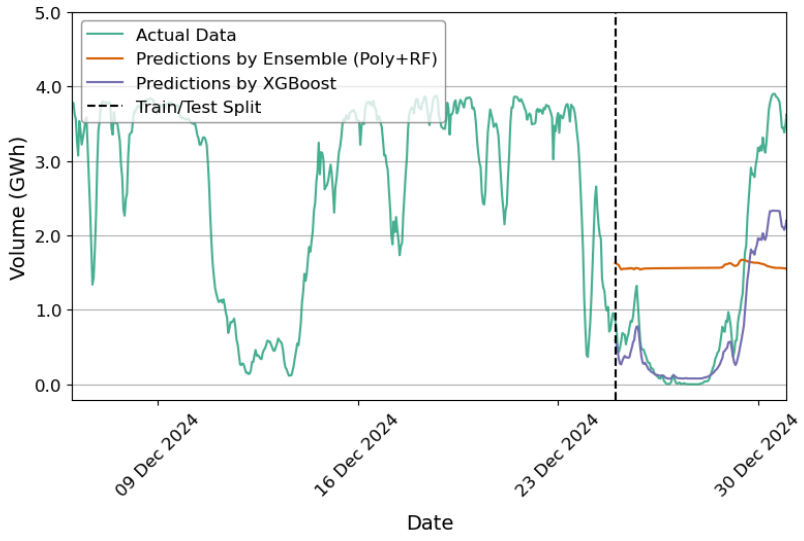
Graph F.1 Hourly Offshore Wind Energy Production: XGBoost with Price Data

Hourly Offshore Wind Energy Production Predictions XGBoost with Price Data



Graph F.2 Hourly Offshore Wind Energy Production: Ensemble and XGBoost

Hourly Offshore Wind Energy Production Predictions (Polynomial+Random Forest and XGBoost)



Graph F.3 Feature Importance Hourly Offshore: XGBoost with Price Data

Feature Importances of XGBoost Offshore Hourly Model with Price Data

